LINEAR PROGRAMMING:
The Solution of Refinery Problems

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Linear Programming:

The Solution of Refinery Problems

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1955

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PREFACE

The material in this book originated in 1953 as a series of intracompany memoranda intended to illustrate the application of linear programming to the solution of refinery problems. In view of the advantages of exchanging this type of information with others, some of the applications were presented as papers at the IBM Petroleum Conference in Endicott, N. Y., during October, 1953, at the Econometric Society Meeting in Washington, D. C., during December, 1953, and at the Society for Advancement of Management Meeting in New York City during January, 1954. The requests for copies of these papers led to the suggestion that the pertinent material be published in book form.

The author is deeply indebted to Professors A. Charnes and W. W. Cooper of the Carnegie Institute of Technology for their instruction in linear programming and for their helpfulness in reviewing this work. Chapter 7 was prepared by Mr. Meyer A. Efroymson of the Process Research Division, Esso Laboratories, of our associate Esso Research and Engineering Company and with his permission has been incorporated into the volume.

GIFFORD H. SYMONDS

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CHAPTER 1

Introduction

Linear programming is a new mathematical method for solving problems concerned with decision-making. It is, therefore, of great interest to management people and their staff advisers. The method is useful in the fields of management science, operations research, and economics. The quantitative aspects of problems such as optimum inventories, production schedules, sales efforts, pricing policies, business expansion, direction of research, evaluation of alternative projects, and economic factors are subject to analysis by linear programming. The qualitative aspects of these problems cannot be handled by linear programming, but with ingenuity many of the qualitative factors can be expressed by quantitative terms. Thus, for example, the quality or appearance of a product, expressed in an arbitrary scale, may be related to its sales demand, price, and production cost.

Linear programming is essentially a method for making systematic selections. The statement of a problem for solution by linear programming involves (1) an objective and (2) a set of restrictions. Thus, in the simplest of examples, let

X_1 = amount of product we will sell,
X_2 = amount of product we will not sell,
100 = maximum capacity for selling product,
\$2 = unit profit for selling product.

Our objective is to make the maximum profit

$$Z = 2X_1 \text{ (max)}$$

and our restriction is the selling capacity

$$X_1 + X_2 = 100.$$

We also require that X_1 and X_2 not be negative:

$$X_1 \geqq 0; X_2 \geqq 0.$$

The solution, of course, is obvious with only a one-dimensional system. Thus

$$X_1 = 100$$
$$X_2 = 0$$

and

$$Z = \$200.$$

The solution mechanism by linear programming is as follows:

Step 1	State restrictions	$X_1 + X_2 = 100 \qquad (X_i \geqq 0).$
Step 2	State objective	$Z = 2X_1$ (max)
Step 3	Determine null case, let	$X_1 = 0.$
	Therefore	$X_2 = 100 = \quad X_1 + X_2$
	and	$Z = \quad 0 = -2X_1 \qquad + Z.$
Step 4	Determine improvement	$\Delta Z/\Delta X_1 = 2.$
Step 5	Make substitution, let	$X_2 = 0.$
	Therefore	$X_1 = 100 = X_1 + \quad X_2$
	and	$Z = 200 = \qquad 2X_2 + Z.$

No further improvement in the objective is possible in this case, and an optimal solution is reached. For such a simple case no mathematical formulation is needed, but for the usual problem with many restrictions and a complicated objective function, the method of linear programming may offer the most efficient solution.

In order to apply linear programming to a problem, it is necessary to state the relationships between the variables as a set of linear equations. The variables must be otherwise independent and must exceed the number of equations. Thus, mathematically,

$$\Sigma_j a_{ij} X_j = b_i \qquad (i = 1,2 \cdots m;\ j = 1,2 \cdots n;\ n > m)$$

where the sum of the coefficient a_{ij} times the value of each variable X_j equals a requirement b_i in the ith equation. These equations are the familiar simultaneous equations such as those used in blending problems. An important difference, of course, is the requirement that the number of variables must exceed the number of equations so that an infinite number of solutions can be obtained and the best one selected. Another requirement is that no quantity can be negative. Thus, mathematically,

$$X_j \geqq 0 \text{ and } b_i \geqq 0.$$

This requirement will assure that only useful answers will be considered.

The solution of the problem by linear programming also requires a directing force or objective which will be maximized (or minimized) during the solution. This is stated mathematically

$$\Sigma_j c_j X_j = \max$$

where c_j is the profit (or cost) per unit of X_j used. The solution is carried out by an iterative process, which at each stage of calculation assigns either zero or positive values to all X_j variables. These values will meet the constraints of the linear equations and will tend to increase the maximizing function at each stage of calculation.

If a solution is possible, a maximum profit case will definitely be reached and other cases with lower profit can also be evaluated.

Linear programming has been applied to the solution of certain refinery problems such as blending available components into required products for maximum profit, optimum crude allocation to several refineries, and optimum inventory and production rate for a seasonal product. Illustrative problems have been set up and solved for these and other problems. Although the values in the illustrative problems are fictional, the real problems are of considerable economic importance and the linear programming solution methods have practical significance.

The most widely applicable method of linear programming is the simplex method, invented by George Dantzig[1] and described by Charnes and Cooper[2] in their *Introduction to Linear Programming*, 1953. This method is suitable for blending and scheduling problems involving different products, operations, and feed supplies where a profit motive can be reduced to terms susceptible to mathematical formulation. Linear programming has also been applied to the solution of game theory problems, both the two-player zero-sum game type[3] and the one-player vs. probability game type.[4] This method is suitable for evaluation of new business ventures or new facilities and for planning optimum inventories against a variable demand. Another application of linear programming is the transportation model,[5] a procedure that applies to optimum distribution of homogeneous material and will solve problems of minimum-cost transportation and maximum-profit allocation.

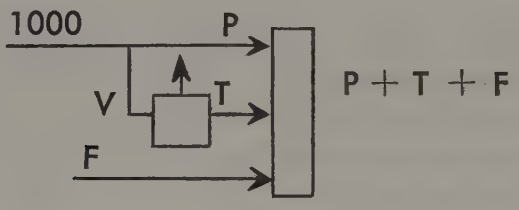
A simplex-method problem involving a 25 by 50 matrix can be solved on a desk computer in about 4 hours per iteration. Solution of the same problem on the IBM 604 multiplier requires about 1 hour per iteration, and on the CPC it is estimated that 25 to 30 minutes per iteration would be required. However, the greatest efficiency may be obtained by using the IBM general solution of the simplex method on its 701 Computer which would require only about 30 seconds per iteration for the 25 by 50 problem. The maximum size simplex method problem currently programmed by IBM is a 50 by 100 matrix. Solutions of the simplex method problems have also been worked out on other high-speed computers such as the CRC and the Univac. The number of iterations required will generally be between m and $2m$, where m is the number of rows in the matrix. Calculations for the transportation model are very simple since only addition and subtraction steps are required. Problems using this model can be solved easily on a desk computer, unless they are very large.

The illustrative material gathered in this volume shows how these problems are set up and solved by the various methods of linear programming. All of the calculations are indicated so that other problems may be set up and solved in similar manner.

CHAPTER 2

A Maximum Profit Fuel Oil Blending Problem Showing the Rudiments of the Mathematical Procedure

A refinery produces, on the average, 1000 gallons per hour of virgin pitch in its crude distillation operation. This pitch may be blended with flux stock to make commercial fuel oil, or it may be sent in whole or in part to a visbreaker unit.



The visbreaker produces an 80 per cent yield of tar which may also be blended with flux stock to make commercial fuel oil. The visbreaking operation is economically break-even if the pitch and the tar are given no value—i.e., the value of the overhead products equals the cost of the operation. The commercial fuel oil brings a realization of 5 cents per gallon, but the flux stock has a cracking value of 8 cents per gallon. This information, together with the viscosity and gravity blending numbers and product specifications, appears in the table below. It is desired to operate for maximum profit.

FUEL OIL BLENDING PROBLEM

	Quan. Avail. GPH	Value CPG	Visc. Bl. No.	Grav. Bl. No.
Pitch	$P = 1000 - V$	0	5	8
Visbreaker feed	V	0	....	....
Tar	$T = 0.8V$	0	11	7
Flux	$F =$ any	8	37	24
Fuel oil	$P + T + F$	5	21 min	12 min

This information may be expressed in the following equations and inequations:

Viscosity requirement $5P + 11T + 37F \geqq 21\,(P + T + F)$.
Gravity requirement $8P + 7T + 24F \geqq 12\,(P + T + F)$.
Material balances $P + V = 1000$
$T = 0.8V$.
Profit will be $Z = 5(P + T + F) - 8F$ cents per hour.

Since there are four relations with four unknowns covering the quality and material requirements, these may be solved directly. The answer will give the quantities of pitch, tar, and flux required to meet the minimum viscosity and gravity specifications of the product. The equations may be restated as follows:

$$(5 - 21)P + (11 - 21)(0.8V) + (37\text{-}21)F = 0$$
$$(8 - 12)P + (7 - 12)(0.8V) + (24\text{-}12)F = 0.$$

$$-2P - V + 2F = 0$$
$$-P - V + 3F = 0.$$

Adding $P + V = 1000$

gives three simultaneous equations, which may be solved:

$$-P + 2F = 1000$$
$$3F = 1000$$
$$F = 333$$
$$P = -333$$
$$V = 1333$$
$$T = 1067.$$

This solution is impossible, since negative quantities are not permissible. The method of simultaneous equations will not guard against negative quantities. In fact, when there are many variables, some negative quantities are almost certain to result. The usual procedure at this point is to look over the requirements to discover what is causing the trouble. Here it is obvious that minimum viscosity and minimum gravity cannot be met at the same time with positive quantities. There must be some give-away in one or both of these qualities. How can maximum profit operation be assured in this situation? This is exactly what linear programming can do. Of course, for small-scale problems, the optimum operation can be surmised or even determined by comparing all the known alternatives; for any size problem it can be calculated unfailingly by linear programming.

One of the basic features of linear programming is that no quantity may be negative. The procedure assures that all requirements will be met and that the maximum

objective will be reached. For a simplified example of linear programming the reader is referred to a paper by A. W. Tucker given at the 1953 ASQC Convention.[10] In this example the requirements are the quality and material balances. The objective is maximum profit.

To solve this problem by the simplex method of linear programming, it is necessary to restate the three simultaneous equations on page 5 as three inequalities:

$$\begin{aligned} 2P + V - 2F &\leq 0 \\ P + V - 3F &\leq 0 \\ P + V &\leq 1000. \end{aligned}$$

The inequalities allow for possible giveaway in viscosity, gravity, or quantity, should this prove to be desirable. These inequalities are changed to equations by adding "giveaway factors." Thus:

$$\begin{aligned} 2P + V - 2F + X_1 &= 0 \\ P + V - 3F + X_2 &= 0 \\ P + V + X_3 &= 1000. \end{aligned}$$

The profit function is expressed in terms of the same variables.

$$\text{Profit} = 5P + 4V - 3F = Z$$

or

$$-5P - 4V + 3F + Z = 0.$$

A basic case is set up, involving the maximum giveaway so that the maximum profit case may be built up in logical order. This basic case may be expressed in tabular form as follows:

TABLEAU I

	P	V	F	X_1	X_2	X_3	Z	Q
X_1	2	1	−2	1	....	....	....	0
X_2	1	1	−3	....	1	....	....	0
X_3	1	1	....	....	....	1	....	1000
Z	−5	−4	3	....	....	....	1	0

It will be noted that this matrix table consists of the coefficients in the preceding equations listed under the appropriate column for each variable. The columns X_1, X_2, X_3, and Z are "identity columns," since the only element in each is a single $+1$. The row corresponding to each of these unity coefficients is known by the same designation as the column, and the quantity Q is the actual value assigned to this variable. The variables X_1, X_2, X_3, and Z are therefore "in the basis," and the variables P, V, and F

are not in the basis and have zero values. The profit function $5P + 4V - 3F$ is shown as the negative in order to include Z in the basis, indicating that these values are only potential at this time.

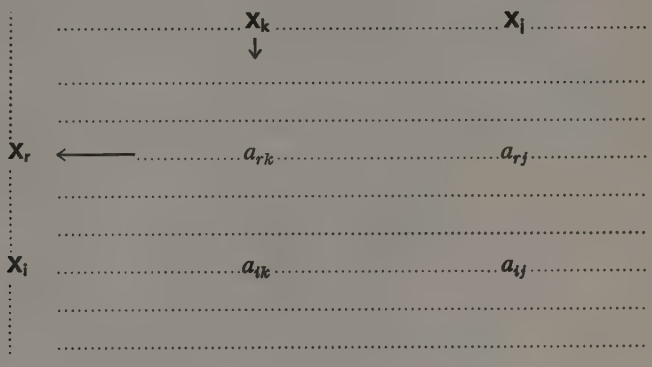
Variable P has the greatest associated profit and should be used first. P must replace X_1 since the smallest positive ratio of Q to P is necessary to avoid negative quantities. The tableau will be transformed to put P into the basis as follows:

TABLEAU II

	P	V	F	X_1	X_2	X_3	Z	Q
Divide equation X_1 by 2 and replace in this position as equation P.								
P	1	1/2	−1	1/2	....	....	....	0
Next subtract this new equation from the X_2 and X_3 equations to obtain the new X_2 and X_3 equations:								
X_2	....	1/2	−2	−1/2	1	....	....	0
X_3	....	1/2	1	−1/2	....	1	....	1000
Then multiply the new P equation by 5 and add to the Z equation to obtain the new Z equation.								
Z	....	3/2	−2	5/2	....	....	1	0

We have now put P into the basis, removing X_1. (Column P is now an identity column and column X_1 is not an identity column.) The quantity associated with P is zero, since, of course, P will not meet the quality requirements unless some flux is used. The desirability of using this flux is shown by the high potential profit under the F column. F must replace X_3 since negative Q to F ratios must be avoided.

Though the method of subtracting equations can be used to transform the tableau, it is simpler to use formulas or "algorithms" for this purpose. The algebraic form of these algorithms is derived as follows:



where

X_r is the row to be replaced,
X_i is any other row vector,
X_k is the column coming into the basis,
and X_j is any other column vector.

	X'_k	X'_j
First divide the equation in row X_r by a_{rk}		
X'_k	$= \frac{X_r}{a_{rk}}$ $(\frac{a_{rk}}{a_{rk}} = 1)$	$\frac{a_{rj}}{a_{rk}}$
Next multiply this equation by a_{ik} and subtract from the equation in row X_i		
X'_i	$(a_{ik} - a_{ik} = 0)$	$a_{ij} - \frac{a_{rj}}{a_{rk}} a_{ik}$

The algorithms may now be stated as follows (Note that the designation of element a_{rj} becomes a'_{kj} since X_k replaces X_r in the basis.):

$$a'_{kj} = \frac{a_{rj}}{a_{rk}},$$

and

$$a'_{ij} = a_{ij} - \frac{a_{rj}}{a_{rk}} a_{ik} \qquad (i \neq k),$$

where

a'_{kj} = any new element in the row replacing the row removed
a'_{ij} = any other new element
a_{ij} = corresponding old element
a_{rj} = old element in the same column and in the row removed
a_{ik} = old element in the same row and in the column entering
a_{rk} = old element in the row removed and in the column entering.

The third tableau may now be prepared by use of the algorithms. The new F row should be calculated first and should replace the old X_3 row.

$$\begin{aligned}
(FV)' &= X_3V/X_3F = 1/2 \div 1 = 1/2 \\
(FF)' &= X_3F/X_3F = 1 \\
(FX_1)' &= X_3X_1/X_3F = -1/2 \div 1 = -1/2 \\
(FX_3)' &= X_3X_3/X_3F = 1 \div 1 = 1 \\
(FQ)' &= X_3Q/X_3F = 1000 \div 1 = 1000
\end{aligned}$$

Only the columns V, F, X_1, X_3, and Q will require transformation since these have ratio values. Column F, of course, will change to an identity column. The calculations for the other columns are given as follows:

$$
\begin{aligned}
(PV)' &= PV - (X_3V/X_3F)(PF) = 1/2 - 1/2(-1) = 1 \\
(X_2V)' &= X_2V - (X_3V/X_3F)(X_2F) = 1/2 - 1/2(-2) = 3/2 \\
(ZV)' &= ZV - (X_3V/X_3F)(ZF) = -3/2 - 1/2(-2) = -1/2 \\
\\
(PX_1)' &= PX_1 - (X_3X_1/X_3F)(PF) = 1/2 + 1/2(-1) = 0 \\
(X_2X_1)' &= X_2X_1 - (X_3X_1/X_3F)(X_2F) = -1/2 + 1/2(-2) = -3/2 \\
(ZX_1)' &= ZX_1 - (X_3X_1/X_3F)(ZF) = 5/2 + 1/2(-2) = 3/2 \\
\\
(PX_3)' &= PX_3 - (X_3X_3/X_3F)(PF) = 0 - 1(-1) = 1 \\
(X_2X_3)' &= X_2X_3 - (X_3X_3/X_3F)(X_2F) = 0 - 1(-2) = 2 \\
(ZX_3)' &= ZX_3 - (X_3X_3/X_3F)(ZF) = 0 - 1(-2) = 2 \\
\\
(PQ)' &= PQ - (X_3Q/X_3F)(PF) = 0 - 1000(-1) = 1000 \\
(X_2Q)' &= X_2Q - (X_3Q/X_3F)(X_2F) = 0 - 1000(-2) = 2000 \\
(ZQ)' &= ZQ - (X_3Q/X_3F)(ZF) = 0 - 1000(-2) = 2000
\end{aligned}
$$

Columns P, X_2, and Z are unchanged. The new tableau may be written:

TABLEAU III

	P	V	F	X_1	X_2	X_3	Z	Q
P	1	1	....	....	....	1	....	1000
X_2	....	3/2	....	−3/2	1	2	....	2000
F	....	1/2	1	−1/2	....	1	....	1000
Z	....	−1/2	....	3/2	....	2	1	2000

This tableau or matrix table pictures the situation where 1000 gph of pitch and 1000 gph of flux are blended to make a fuel oil meeting the minimum viscosity specification ($X_1 = 0$) and giving away 4 points on gravity. No pitch is being visbroken, and the profit is 2000 cents per hour ($480 per day). This is not the maximum profit case, however, since some potential profit is shown in the V column. V should enter the basis, replacing P, since the Q to V ratio of 1000 is the smallest.

The calculations are made as follows:

$$
\begin{aligned}
(VP)' &= 1/1 = 1 \\
(VV)' &= 1/1 = 1 \\
(VX_3)' &= 1/1 = 1 \\
(VQ)' &= 1000/1 = 1000
\end{aligned}
$$

$$
\begin{aligned}
(X_2P)' &= 0 - 1(\ 3/2) = -3/2 \\
(FP)' &= 0 - 1(\ 1/2) = -1/2 \\
(ZP)' &= 0 - 1(-1/2) = 1/2
\end{aligned}
$$

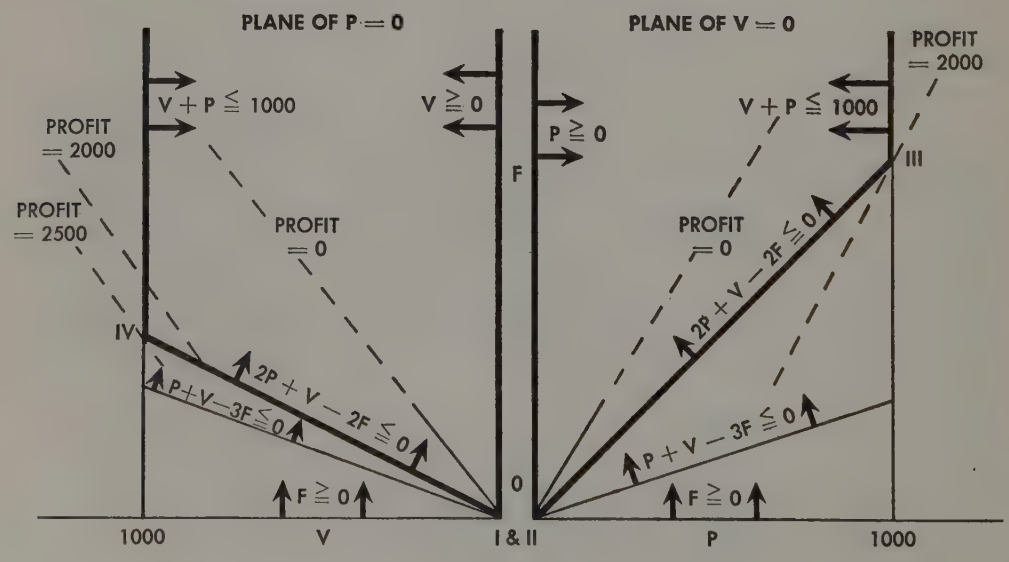
$$
\begin{aligned}
(X_2X_3)' &= 2 - 1(\ 3/2) = 1/2 \\
(FX_3)' &= 1 - 1(\ 1/2) = 1/2 \\
(ZX_3)' &= 2 - 1(-1/2) = 5/2
\end{aligned}
$$

$$
\begin{aligned}
(X_2Q)' &= 2000 - 1000(\ 3/2) = 500 \\
(FQ)' &= 1000 - 1000(\ 1/2) = 500 \\
(ZQ)' &= 2000 - 1000(-1/2) = 2500
\end{aligned}
$$

Column V is transformed into an identity column. Columns F, X_1, X_2, and Z are unchanged. The new tableau may be written as follows:

TABLEAU IV

	P	V	F	X_1	X_2	X_3	Z	Q
V	1	1	....	....	....	1	....	1000
X_2	−3/2	....	....	−3/2	1	1/2	....	500
F	−1/2	....	1	−1/2	....	1/2	....	500
Z	1/2	....	....	3/2	....	5/2	1	2500



The above case represents the maximum possible profit in this operation since there is no potential profit (no negative value in the Z row) to be obtained by using any other variable in the basis. This case includes visbreaking all of the pitch (1000 gph) and blending the tar (800 gph) with 500 gph of flux. The minimum viscosity specification is exactly met, but there is a giveaway of about 1.5 points in gravity. The maximum profit is 2500 cents per hour ($600 per day).

A geometrical representation of this solution may be seen at the bottom of page 10.

The three-dimensional diagram indicates the constraints due to viscosity, gravity, and availability of pitch, with the constant profit lines shown for each plane. The quantity of flux stock is minimized by the maximal value of the profit functional when all available pitch is used. The gravity constraint is not limiting and could have been eliminated from the set.

CHAPTER 3

A Maximum Profit Running Plan Problem

Running plans that will assure maximum possible profit can be prepared by the simplex method of linear programming. An example pertaining to a special form of the running plan problem was prepared by Charnes, Cooper, and Mellon[6] in their integrated oil company study. An example of the general form of preparing a running plan is given herewith.

A refinery has four different crudes available in stated quantities and wishes to meet given requirements for four different products. One of the crudes can be run alternately either for fuel products or for lube distillate. The product yields and the profit for each crude operation are given. The objective is to make a selection of crude oils which will meet product requirements with maximum possible profit.

TABLEAU I

	A	B	C_1	C_2 ↓	D	X_A	X_B	X_C	X_D	X_G	X_H	X_L	X_F	Z	Q
X_A	1	....	....	....	....	1	....	....	....	....	....	....	....	....	100
X_B	....	1	....	....	....	....	1	....	....	....	....	....	....	....	100
X_C	....	....	1	1	....	....	....	1	....	....	....	....	....	....	200
X_D	....	....	....	....	1	....	....	....	1	....	....	....	....	....	100
X_G	0.6	0.5	0.4	0.4	0.3	....	....	....	....	1	....	....	....	....	170
X_H	0.2	0.2	0.3	0.1	0.3	....	....	....	....	....	1	....	....	....	85
X_L ←	....	....	....	0.2	....	....	....	....	....	....	....	1	....	....	20
X_F	0.1	0.2	0.2	0.2	0.3	....	....	....	....	....	....	....	1	....	85
Z	−10	−20	−15	−25	−7	....	....	....	....	....	....	....	....	1	0

Tableau I contains a complete statement of the problem. The four crudes, A, B, C, and D, are available in quantities Q of 100, 100, 200, and 100 Mbpw respectively. The desired products, G, H, L, and F, have requirements of 170, 85, 20, and 85 Mbpw respectively. The yields for each crude are shown as decimal fractions and the negative profit as cents per barrel listed in the column for each crude. The vectors X_A, X_B, X_C, and X_D represent the respective crudes not run, and the vectors X_G, X_H,

X_L, and X_F represent the respective product requirements not met. The matrix table is a basic feasible solution and indicates that no crudes are being run and no product requirements are met. The profit Z is 0.

The next feasible solution (No. 2) of the problem is selected by introducing the column showing the largest negative Z value as a row in the matrix. This is column C_2 with a Z value of -25. The row in the first matrix which is to be replaced is determined by the minimum ratio of Q to C_2, using only positive values. This is row X_L with a ratio of $20/0.2 = 100$. Column C_2 and row X_L are marked with arrows. This means that crude operation C_2 will be run to meet all of the lube distillate requirements. This is done preferentially because it is the most profitable operation. The new matrix is set up as follows, with identity columns for the common vectors.

	A	B	C_1	C_2	D	X_A	X_B	X_C	X_D	X_G	X_H	X_L	X_F	Z	Q
X_A	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....
X_B	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....
X_C	....	....	....	....	....	....	....	1	....	....	....	....	....	....	....
X_D	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....
X_G	....	....	....	....	....	....	....	....	....	1	....	....	....	....	....
X_H	....	....	....	....	....	....	....	....	....	....	1	....	....	....	....
C_2	....	....	....	1	....	....	....	....	....	....	....	5	....	....	100
X_F	....	....	....	....	....	....	....	....	....	....	....	....	1	....	....
Z	....	....	....	....	....	....	....	....	....	....	....	....	....	1	....

Row C_2 is also completed by dividing each element in the X_L row of matrix 1 by X_LC_2, thus obtaining $0.2/0.2 = 1$, $1/0.2 = 5$, and $20/0.2 = 100$ respectively for the C_2, X_L, and Q elements in matrix 2.

The columns C_2, X_L, and Q now require correction. Since column C_2 is an identity vector, all other elements in this column are 0. Column X_L is corrected as follows:

$$(X_AX_L)_2 = (X_AX_L)_1 - (C_2X_L)_2(X_AC_2)_1 = 0 - 5 \times 0 = 0$$

$$(X_BX_L)_2 = (X_BX_L)_1 - (C_2X_L)_2(X_BC_2)_1 = 0 - 5 \times 0 = 0$$

$$(X_CX_L)_2 = (X_CX_L)_1 - (C_2X_L)_2(X_CC_2)_1 = 0 - 5 \times 1 = -5$$

$$(X_DX_L)_2 = (X_DX_L)_1 - (C_2X_L)_2(X_DC_2)_1 = 0 - 5 \times 0 = 0$$

$$(X_GX_L)_2 = (X_GX_L)_1 - (C_2X_L)_2(X_GC_2)_1 = 0 - 5 \times 0.4 = -2$$

$$(X_HX_L)_2 = (X_HX_L)_1 - (C_2X_L)_2(X_HC_2)_1 = 0 - 5 \times 0.1 = -0.5$$

$$(X_FX_L)_2 = (X_FX_L)_1 - (C_2X_L)_2(X_FC_2)_1 = 0 - 5 \times 0.2 = -1$$

$$(ZX_L)_2 = (ZX_L)_1 - (C_2X_L)_2(ZC_2)_1 = 0 - 5 \times (-25) = 125$$

and column Q is corrected as follows:

$$(X_AQ)_2 = (X_AQ)_1 - (C_2Q)_2(X_AC_2)_1 = 100 - 100 \times 0 = 100$$
$$(X_BQ)_2 = (X_BQ)_1 - (C_2Q)_2(X_BC_2)_1 = 100 - 100 \times 0 = 100$$
$$(X_CQ)_2 = (X_CQ)_1 - (C_2Q)_2(X_CC_2)_1 = 200 - 100 \times 1 = 100$$
$$(X_DQ)_2 = (X_DQ)_1 - (C_2Q)_2(X_DC_2)_1 = 100 - 100 \times 0 = 100$$
$$(X_GQ)_2 = (X_GQ)_1 - (C_2Q)_2(X_GC_2)_1 = 170 - 100 \times 0.4 = 130$$
$$(X_HQ)_2 = (X_HQ)_1 - (C_2Q)_2(X_HC_2)_1 = 85 - 100 \times 0.1 = 75$$
$$(X_FQ)_2 = (X_FQ)_1 - (C_2Q)_2(X_FC_2)_1 = 85 - 100 \times 0.2 = 65$$
$$(ZQ)_2 = (ZQ)_1 - (C_2Q)_2(ZC_2)_1 = 0 - 100(-25) = 2500$$

All other columns remain unchanged. The new matrix for feasible solution No. 2 is completed as follows:

TABLEAU II

	A	B ↓	C_1	C_2	D	X_A	X_B	X_C	X_D	X_G	X_H	X_L	X_F	Z	Q
X_A	1	....	....	....	....	1	....	....	....	....	....	....	....	....	100
X_B ←	....	1	....	....	....	....	1	....	....	....	....	....	....	....	100
X_C	....	....	1	....	....	....	....	1	....	....	....	–5	....	....	100
X_D	....	....	....	....	1	....	....	....	1	....	....	....	....	....	100
X_G	0.6	0.5	0.4	....	0.3	....	....	....	....	1	....	–2	....	....	130
X_H	0.2	0.2	0.3	....	0.3	....	....	....	....	....	1	–0.5	....	....	75
C_2	....	....	....	1	....	....	....	....	....	....	....	5	....	....	100
X_F	0.1	0.2	0.2	....	0.3	....	....	....	....	....	....	–1	1	....	65
Z	–10	–20	–15	....	–7	....	....	....	....	....	....	125	....	1	2500

As the tableau shows, the profit for running 100 Mbpw of crude *C* to meet the 20 Mbpw of lube distillate requirements is 2500 (1000/100) = $25,000 per week. Remaining are 400 Mbpw of available crudes and requirements for *G*, *H*, and *F* products of 130, 75, and 65 Mbpw respectively. The next feasible solution is determined by inspection of the matrix table. Column *B* contains the largest negative *Z* value of —20, and row X_B shows the smallest *Q* to *B* ratio of 100/1 = 100. The new matrix, Tableau III, is set up as before.

As shown, the profit for running 200 Mbpw of crudes *B* and *C* is 4500 (1000/100) = $45,000 per week. Remaining are 300 Mbpw of crudes *A*, *C*, and *D* and requirements for *G*, *H*, and *F* products of 80, 55, and 45 Mbpw respectively. Column C_1 is introduced next ($ZC_1 = -15$) and row X_C is removed ($Q/C_1 = 100$). Tableau IV, the new matrix, is set up as before.

TABLEAU III

	A	B	C_1 ↓	C_2	D	X_A	X_B	X_C	X_D	X_G	X_H	X_L	X_F	Z	Q
X_A	1	....	....	....	....	1	....	....	....	....	....	....	....	....	100
B	....	1	....	....	....	....	1	....	....	....	....	....	....	....	100
X_C ←	....	....	1	....	....	....	....	1	....	....	....	−5	....	....	100
X_D	....	....	....	....	1	....	....	....	1	....	....	....	....	....	100
X_G	0.6	....	0.4	....	0.3	....	−0.5	....	....	1	....	−2	....	....	80
X_H	0.2	....	0.3	....	0.3	....	−0.2	....	....	....	1	−0.5	....	....	55
C_2	....	....	....	1	....	....	....	....	....	....	....	5	....	....	100
X_F	0.1	....	0.2	....	0.3	....	−0.2	....	....	....	....	−1	1	....	45
Z	−10	....	−15	....	−7	....	20	....	....	....	....	125	....	1	4500

TABLEAU IV

	A ↓	B	C_1	C_2	D	X_A	X_B	X_C	X_D	X_G	X_H	X_L	X_F	Z	Q
X_A	1	....	....	....	....	1	....	....	....	....	....	....	....	....	100
B	....	1	....	....	....	....	1	....	....	....	....	....	....	....	100
C_1	....	....	1	....	....	....	....	1	....	....	....	−5	....	....	100
X_D	....	....	....	....	1	....	....	....	1	....	....	....	....	....	100
X_G ←	0.6	....	....	....	0.3	....	−0.5	−0.4	....	1	....	....	....	....	40
X_H	0.2	....	....	....	0.3	....	−0.2	−0.3	....	....	1	1	....	....	25
C_2	....	....	....	1	....	....	....	....	....	....	....	5	....	....	100
X_F	0.1	....	....	....	0.3	....	−0.2	−0.2	....	....	....	....	1	....	25
Z	−10	....	....	....	−7	....	20	15	....	....	....	50	....	1	6000

As shown, the profit for running 300 Mbpw of crudes *B* and *C* is 6000 (1000/100) = $60,000 per week. Remaining are 200 Mbpw of crudes *A* and *D* and requirements for *G*, *H*, and *F* products of 40, 25, and 25 Mbpw respectively. Column *A* is introduced next ($ZA = -10$), and row X_G is removed ($Q/A = 67$). The new matrix is set up as before.

As Tableau V, page 16, shows, the profit for running 367 Mbpw of crudes *A*, *B*, and *C* is 6667 (1000/100) = $66,670 per week. Remaining are 133 Mbpw of crudes *A* and *D* and requirements for *H* and *F* products of 12 and 18 Mbpw respectively. Column *D* is introduced ($ZD = -2$), and row X_H is removed ($Q/D = 58$). The new matrix is set up as before.

Since there are no negative values in the *Z* row, Tableau VI, this case represents a

TABLEAU V

	A	B	C_1	C_2	D ↓	X_A	X_B	X_C	X_D	X_G	X_H	X_L	X_F	Z	Q
X_A	....	....	....	....	−0.5	1	5/6	2/3	....	−5/3	....	....	....	....	33
B	....	1	....	....	....	....	1	....	....	....	....	....	....	....	100
C_1	....	....	1	....	....	....	....	1	....	....	....	−5	....	....	100
X_D	....	....	....	....	1	....	....	....	1	....	....	....	....	....	100
A	1	....	....	....	0.5	....	−5/6	−2/3	....	5/3	....	....	....	....	67
X_H ←	....	....	....	....	0.2	....	−1/30	−1/6	....	−1/3	1	1	....	....	12
C_2	....	....	....	1	....	....	....	....	....	....	....	5	....	....	100
X_F	....	....	....	....	0.25	....	−7/60	−2/15	....	−1/6	....	....	1	....	18
Z	....	....	....	....	−2	....	35/3	25/3	....	50/3	....	50	....	1	6667

TABLEAU VI

	A	B	C_1	C_2	D	X_A	X_B	X_C	X_D	X_G	X_H	X_L	X_F	Z	Q
X_A	....	....	....	....	....	1	3/4	1/4	....	−2.5	2.5	2.5	....	....	62
B	....	1	....	....	....	....	1	....	....	....	....	....	....	....	100
C_1	....	....	1	....	....	....	....	1	....	....	....	−5	....	....	100
X_D	....	....	....	....	....	....	1/6	5/6	1	5/3	−5	−5	....	....	42
A	1	....	....	....	....	....	−3/4	−1/4	....	2.5	−2.5	−2.5	....	....	38
D	....	....	....	....	1	....	−1/6	−5/6	....	−5/3	5	5	....	....	58
C_2	....	....	....	1	....	....	....	....	....	....	....	5	....	....	100
X_F	....	....	....	....	....	....	−3/40	3/40	....	1/4	−5/4	−5/4	1	....	4
Z	....	....	....	....	....	....	34/3	20/3	....	40/3	10	60	....	1	6783

maximum profit feasible solution. The profit is 6783(1000/100) = $67,830 per week. The following table shows the disposition of available crudes in this solution.

		Crudes Mbpw						
		A	B	C_1	C_2	D	Product Not Made	Total Required
Products	*G*	22	50	40	40	18	0	170
Mbpw	*H*	7	20	30	10	18	0	85
	L	....	....	....	20	....	0	20
	F	4	20	20	20	17	4	85
Crude not used		62	0	0	0	42		
Total crude		100	100	100	100	100		

With the given crude slate the requirements for products *G*, *H*, and *L* are met, but it is not economical to meet quite all of the requirements for product *F*. Of course, more of crude *D* could be used to displace the other crudes in order to increase the yield of product *F* with some loss of profit. Other methods would include the use of different crudes having high *F* yield, the purchase of product *F* for resale, and reducing the percentage of product *F* required.

CHAPTER 4

A Problem in Blending Aviation Gasoline for Maximum Profit

Blending problems are very well adaptable for solution by the simplex method of linear programming. Some studies on blending aviation gasoline have been made available by Charnes, Cooper, and Mellon[7]. The example given herewith gives the general solution and illustrates some simplicity in handling the data.

It is desired to blend five available refinery streams for the requirements of Grade 100/130 and Grade 115/145 aviation gasoline to make maximum profit. A study of the past shipments indicates that only three inspections are critical for each grade, namely 1C +4.6 lean octane (performance number, or PN), 3C +4.6 rich octane (PN), and Reid vapor pressure (RVP). The inspections used in this study are linear blending values (BL) including tolerances. Neglecting the noncritical inspections is one way of simplifying the blending problem. Complete inspections for the final blends can be calculated to be sure that product quality will be satisfactory in all respects. The critical inspections, cost, and availability of the refinery streams are shown as follows:

Stream	BL PN 1C +4.6	BL PN 3C +4.6	BL RVP	Cost cpg	Avail bpd
No. 1	122.0	158.3	3.0	17.7	750
No. 2	109.0	145.0	4.0	16.0	800
No. 3	97.7	110.0	5.0	14.5	857
No. 4	90.0	125.0	7.0	13.0	1000
No. 5	114.0	145.0	19.0	11.0	Any

In view of its low cost and high quality, it is obvious that the maximum amount of stream 5 should be used in the blends as limited by the RVP specification of 7 on each grade. Thus, the RVP inspection and stream 5 can be eliminated from the prob-

lem by blending streams 1 to 4 up to 7 RVP with stream 5. This procedure is another way of simplifying the blending problem. The resulting preliminary blends are shown as follows:

Component	BL PN 1C +4.6	BL PN 3C +4.6	BL RVP	Cost cpg	Avail bpd
A	120.0	155.0	7	16.0	1000
B	110.0	145.0	7	15.0	1000
C	100.0	115.0	7	14.0	1000
D	90.0	125.0	7	13.0	1000

The specifications, refinery prices, and requirements of the products are shown as follows:

Product	BL PN 1C +4.6	BL PN 3C +4.6	BL RVP	Price cpg	Required bpd
M	100	130	7	16.0	2000
N	115	145	7	18.0	2000

The linear equations are developed from this information.

The quantity of each component used plus the amount of product not made equals the requirement.

$$A_M + B_M + C_M + D_M + X_M = M = 2000$$
$$A_N + B_N + C_N + D_N + X_N = N = 2000$$

The quantity of each component used plus the amount not used equals the availability.

$$A_M + A_N + X_A = A = 1000$$
$$B_M + B_N + X_B = B = 1000$$
$$C_M + C_N + X_C = C = 1000$$
$$D_M + D_N + X_D = D = 1000$$

The quality relationships are expressed as inequalities to allow for possible product quality giveaway.

$$120\,A_M + 110\,B_M + 100\,C_M + 90\,D_M \geqq 100\,(M - X_M)$$
$$155\,A_M + 145\,B_M + 115\,C_M + 125\,D_M \geqq 130\,(M - X_M)$$
$$120\,A_N + 110\,B_N + 100\,C_N + 90\,D_N \geqq 115\,(N - X_N)$$
$$155\,A_N + 145\,B_N + 115\,C_N + 125\,D_N \geqq 145\,(N - X_N)$$

These relationships can be expressed as differential inequalities as follows:

$$\begin{aligned}
20\,A_M + 10\,B_M &- 10\,D_M \geqq 0\\
25\,A_M + 15\,B_M - 15\,C_M &- 5\,D_M \geqq 0\\
5\,A_N - 5\,B_N - 15\,C_N &- 25\,D_N \geqq 0\\
10\,A_N - 30\,C_N &- 20\,D_N \geqq 0.
\end{aligned}$$

These inequalities can be simplified and expressed as equations:

$$\begin{aligned}
-2\,A_M - B_M + D_M + X_P &= 0\\
-5\,A_M - 3\,B_M + 3\,C_M + D_M + X_Q &= 0\\
-A_N + B_N + 3\,C_N + 5\,D_N + X_R &= 0\\
-A_N + 3\,C_N + 2\,D_N + X_S &= 0.
\end{aligned}$$

The equation for profit can be written as follows:

$$Z = 2\,A_N + B_M + 3\,B_N + 2\,C_M + 4\,C_N + 3\,D_M + 5\,D_N.$$

Since it is desired to maximize this function, the negative value is used in the matrix.

The basic feasible solution and the ten substituted cases are shown on pages 22-24. The maximum profit feasible solution can be summarized as follows:

Component	M 100/130	N 115/145	Not Used	Avail
A	0	1000	0	1000
B	750	250	0	1000
C	500	250	250	1000
D	750	0	250	1000
Not Made	0	500	....	....
Required	2000	2000	....	....

The maximum profit for this case is 7750(42/100) = $3255 per day. The requirement for Grade 100/130 is met, and preliminary blends A and B are used entirely; it is not economical, however, to meet all of the requirement for Grade 115/145 with the available refinery streams. Part of preliminary blends C and D are not used and would be available for motor gasoline.

The critical inspections of the final blends are determined as shown in the table at the top of page 21.

The product quality giveaway of 1.7 PN on Grade 115/145 is the best arrangement for maximum profit with the existing situation. Refinery process changes which would economically increase the ratio of 1C to 3C rating of the blending stocks slightly would be worth investigating to avoid this giveaway.

CRITICAL INSPECTIONS OF THE FINAL BLENDS

	Product M(100/130)			Product N(115/145)		
		Inspect BL PN			Inspect BL PN	
Component	% in Blend	1C +4.6	3C +4.6	% in Blend	1C +4.6	3C +4.6
A	....	....	....	66.7	120.0	155.0
B	37.5	110.0	145.0	16.7	110.0	145.0
C	25.0	100.0	115.0	16.7	100.0	115.0
D	37.5	90.0	125.0	....	....	....
	100	100.0	130.0	100	115.0	146.7

(Note $X_g = 250$; $\triangle 3C = 250(10/1500) = 1.7$ PN)

If it is necessary to make the maximum amount of Grade 115/145 regardless of cost, the highest octane components should be blended.

Component	BL PN 1C +4.6	BL PN 3C +4.6	BL RVP	Cost cpg	Avail bpd
A	120.0	155.0	7	16.0	1000
B	110.0	145.0	7	15.0	1000
Max. 115/145	115.0	150.0	7	15.5	2000
Excess	95.0	120.0	7	13.5	2000

In this case the requirement of 2000 bpd for Grade 115/145 is met with 5 PN 3C +4.6 octane giveaway. The excess blending stock of 2000 bpd must be returned to motor gasoline since it will not make any Grade 100/130 because of low octane quality. The profit with this method is 2000 (18.0 — 15.5) (42/100) = $2100 per day. This, of course, is far below the maximum profit of $3255 per day for the case determined by linear programming.

It should be pointed out that it is possible to convert all 4000 bpd of the preliminary blends A to D into 1333 bpd of Grade 115/145 and 2667 bpd of Grade 100/130 aviation gasoline. Since the requirement for Grade 100/130 is only 2000 bpd, the excess of 667 bpd would have to be returned to motor gasoline. The profit in this case would be [1333 (18 — 15.5) + 2000 (16 — 14.0)] (42/100) = $3080 per day. This is less than the maximum profit of $3255 per day for the case determined by linear programming.

Basic Feasible Solution

Tableau I

	A_M	A_N	B_M	B_N	C_M	C_N	D_M	D_N ↓	X_A	X_B	X_C	X_D	X_M	X_N	X_P	X_Q	X_R	X_S	Z	Q
X_A	1	1	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	1000
X_B	....	....	1	1	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	1000
X_C	....	....	....	....	1	1	....	....	....	....	1	....	....	....	....	....	....	....	....	1000
X_D	....	....	....	....	....	....	1	1	....	....	....	1	....	....	....	....	....	....	....	1000
X_M	1	....	1	....	1	....	1	....	....	....	....	....	1	....	....	....	....	....	....	2000
X_N	....	1	....	1	....	1	....	1	....	....	....	....	....	1	....	....	....	....	....	2000
X_P	—2	....	—1	....	0	....	1	....	....	....	....	....	....	....	1	....	....	....	....	0
X_Q	—5	....	—3	....	3	....	1	....	....	....	....	....	....	....	....	1	....	....	....	0
X_R ←	....	—1	....	1	....	3	....	5	....	....	....	....	....	....	....	....	1	....	....	0
X_S	....	—1	....	0	....	3	....	2	....	....	....	....	....	....	....	....	....	1	....	0
Z	0	—2	—1	—3	—2	—4	—3	—5	....	....	....	....	....	....	....	....	....	....	1	0

Tableau II

	A_M	A_N	B_M	B_N	C_M	C_N	D_M ↓	D_N	X_A	X_B	X_C	X_D	X_M	X_N	X_P	X_Q	X_R	X_S	Z	Q
X_A	1	1	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	1000
X_B	....	....	1	1	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	1000
X_C	....	....	....	....	1	1	....	....	....	....	1	....	....	....	....	....	....	....	....	1000
X_D	....	1/5	....	—1/5	....	—3/5	1	....	....	....	....	1	....	....	....	....	—1/5	....	....	1000
X_M	1	....	1	....	1	....	1	....	....	....	....	....	1	....	....	....	....	....	....	2000
X_N	....	6/5	....	4/5	....	2/5	....	....	....	....	....	....	....	1	....	....	—1/5	....	....	2000
X_P ←	—2	....	—1	....	....	....	1	....	....	....	....	....	....	....	1	....	....	....	....	....
X_Q	—5	....	—3	....	3	....	1	....	....	....	....	....	....	....	....	1	....	....	....	....
D_N	....	—1/5	....	1/5	....	3/5	....	1	....	....	....	....	....	....	....	....	1/5	....	....	....
X_S	....	—3/5	....	—2/5	....	9/5	....	....	....	....	....	....	....	....	....	....	—2/5	1	....	....
Z	....	—3	—1	—2	—2	—1	—3	....	....	....	....	....	....	....	....	....	1	....	1	....

Tableau III

	A_M ↓	A_N	B_M	B_N	C_M	C_N	D_M	D_N	X_A	X_B	X_C	X_D	X_M	X_N	X_P	X_Q	X_R	X_S	Z	Q
X_A	1	1	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	1000
X_B	....	....	1	1	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	1000
X_C	....	....	....	....	1	1	....	....	....	....	1	....	....	....	....	....	....	....	....	1000
X_D ←	2	1/5	1	—1/5	....	—3/5	....	....	....	....	....	1	....	....	—1	....	—1/5	....	....	1000
X_M	3	....	2	....	1	....	....	....	....	....	....	....	1	....	—1	....	....	....	....	2000
X_N	....	6/5	....	4/5	....	2/5	....	....	....	....	....	....	....	1	....	....	—1/5	....	....	2000
D_M	—2	....	—1	....	....	....	1	....	....	....	....	....	....	....	1	....	....	....	....	....
X_Q	—3	....	—2	....	3	....	....	....	....	....	....	....	....	....	—1	1	....	....	....	....
D_N	....	—1/5	....	1/5	....	3/5	....	1	....	....	....	....	....	....	....	....	1/5	....	....	....
X_S	....	—3/5	....	—2/5	....	9/5	....	....	....	....	....	....	....	....	....	....	—2/5	1	....	....
Z	—6	—3	—4	—2	—2	—1	....	....	....	....	....	....	....	....	3	....	1	....	1	....

Tableau IV

	A_M	A_N	B_M	B_N	C_M	C_N ↓	D_M	D_N	X_A	X_B	X_C	X_D	X_M	X_N	X_P	X_Q	X_R	X_S	Z	Q
X_A	....	9/10	—1/2	1/10	....	3/10	....	....	1	....	....	—1/2	....	....	1/2	....	1/10	....	....	500
X_B	....	....	1	1	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	1000
X_C	....	....	....	....	1	1	....	....	....	....	1	....	....	....	....	....	....	....	....	1000
A_M	1	1/10	1/2	—1/10	....	—3/10	....	....	....	....	....	1/2	....	....	—1/2	....	—1/10	....	....	500
X_M	....	—3/10	1/2	3/10	1	9/10	....	....	....	....	....	—3/2	1	....	1/2	....	3/10	....	....	500
X_N	....	6/5	....	4/5	....	2/5	....	....	....	....	....	....	....	1	....	....	—1/5	....	....	2000
D_M	....	1/5	....	—1/5	....	—3/5	1	....	....	....	....	1	....	....	....	....	—1/5	....	....	1000
X_Q	....	3/10	—1/2	—3/10	3	—9/10	....	....	....	....	....	3/2	....	....	—5/2	1	—3/10	....	....	1500
D_N	....	—1/5	....	1/5	....	3/5	....	1	....	....	....	....	....	....	....	....	1/5	....	....	....
X_S ←	....	—3/5	....	—2/5	....	9/5	....	....	....	....	....	....	....	....	....	....	—2/5	1	....	....
Z	....	—12/5	—1	—13/5	—2	—14/5	....	....	....	....	....	3	....	....	....	....	2/5	....	1	3000

Tableau V

	A_M	A_N ↓	B_M	B_N	C_M	C_N	D_M	D_N	X_A	X_B	X_C	X_D	X_M	X_N	X_P	X_Q	X_R	X_S	Z	Q
X_A ←	....	1	—1/2	1/6	....	....	....	....	1	....	....	—1/2	....	....	1/2	....	1/6	—1/6	....	500
X_B	....	....	1	1	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	1000
X_C	....	1/3	....	2/9	1	....	....	....	....	....	1	....	....	....	....	....	2/9	—5/9	....	1000
A_M	1	....	1/2	—1/6	....	....	....	....	....	....	....	1/2	....	....	—1/2	....	—1/6	1/6	....	500
X_M	....	....	1/2	1/2	1	....	....	....	....	....	....	—3/2	1	....	1/2	....	1/2	—1/2	....	500
X_N	....	4/3	....	8/9	....	....	....	....	....	....	....	....	....	1	....	....	—1/9	—2/9	....	2000
D_M	....	....	....	—1/3	....	....	1	....	....	....	....	1	....	....	....	....	—1/3	1/3	....	1000
X_Q	....	....	—1/2	—1/2	3	....	....	....	....	....	....	3/2	....	....	—5/2	1	—1/2	1/2	....	1500
D_N	....	....	....	1/3	....	....	....	1	....	....	....	....	....	....	....	....	1/3	—1/3	....	....
C_N	....	—1/3	....	—2/9	....	1	....	....	....	....	....	....	....	....	....	....	—2/9	5/9	....	....
Z	....	—10/3	—1	—29/9	—2	....	....	....	....	....	....	3	....	....	....	....	—2/9	14/9	1	3000

Tableau VI

	A_M	A_N	B_M ↓	B_N	C_M	C_N	D_M	D_N	X_A	X_B	X_C	X_D	X_M	X_N	X_P	X_Q	X_R	X_S	Z	Q
A_N	....	1	—1/2	1/6	....	....	....	....	1	....	....	—1/2	....	....	1/2	....	1/6	—1/6	....	500
X_B ←	....	....	1	1	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	1000
X_C	....	....	1/6	1/6	1	....	....	....	—1/3	....	1	1/6	....	....	—1/6	....	1/6	—1/2	....	833
A_M	1	....	1/2	—1/6	....	....	....	....	....	....	....	1/2	....	....	—1/2	....	—1/6	1/6	....	500
X_M	....	....	1/2	1/2	1	....	....	....	....	....	....	—3/2	1	....	1/2	....	1/2	—1/2	....	500
X_N	....	....	2/3	2/3	....	....	....	....	—4/3	....	....	2/3	....	1	—2/3	....	—1/3	....	....	1333
D_M	....	....	....	—1/3	....	....	1	....	....	....	....	1	....	....	....	....	—1/3	1/3	....	1000
X_Q	....	....	—1/2	—1/2	3	....	....	....	....	....	....	3/2	....	....	—5/2	1	—1/2	1/2	....	1500
D_N	....	....	....	1/3	....	....	....	1	....	....	....	....	....	....	....	....	1/3	—1/3	....	....
C_N	....	....	—1/6	—1/6	....	1	....	....	1/3	....	....	—1/6	....	....	1/6	....	—1/6	1/2	....	167
Z	....	....	—8/3	—8/3	—2	....	....	....	10/3	....	....	4/3	....	....	5/3	....	1/3	1	1	4667

Tableau VII

	A_M	A_N	B_M	B_N	C_M ↓	C_N	D_M	D_N	X_A	X_B	X_C	X_D	X_M	X_N	X_P	X_Q	X_R	X_S	Z	Q
A_N	....	1	....	2/3	....	....	....	....	1	1/2	....	—1/2	....	....	1/2	....	1/6	—1/6	....	1000
B_M	....	....	1	1	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	1000
X_C	....	....	....	....	1	....	....	....	—1/3	—1/6	1	1/6	....	....	—1/6	....	1/6	—1/2	....	667
A_M	1	....	....	—2/3	....	....	....	....	....	—1/2	....	1/2	....	....	—1/2	....	—1/6	1/6	....	....
X_M ←	....	....	....	....	1	....	....	....	....	—1/2	....	—3/2	1	....	1/2	....	1/2	—1/2	....	....
X_N	....	....	....	....	....	....	....	....	—4/3	—2/3	....	2/3	....	1	—2/3	....	—1/3	....	....	667
D_M	....	....	....	—1/3	....	....	1	....	....	....	....	1	....	....	....	....	—1/3	1/3	....	1000
X_Q	....	....	....	....	3	....	....	....	....	1/2	....	3/2	....	....	—5/2	1	—1/2	1/2	....	2000
D_N	....	....	....	1/3	....	....	1	....	....	....	....	....	....	....	....	....	1/3	—1/3	....	....
C_N	....	....	....	....	1	....	....	....	1/3	1/6	....	—1/6	....	....	1/6	....	—1/6	1/2	....	333
Z	....	....	....	....	—2	....	....	....	10/3	8/3	....	4/3	....	....	5/3	....	1/3	1	1	7333

Tableau VIII

	A_M	A_N	B_M	B_N	C_M	C_N	D_M	D_N	X_A	X_B	X_C	X_D ↓	X_M	X_N	X_P	X_Q	X_R	X_S	Z	Q
A_N	....	1	....	2/3	....	....	....	....	1	1/2	....	—1/2	....	....	1/2	....	1/6	—1/6	....	1000
B_M	....	....	1	1	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	1000
X_C	....	....	....	....	....	....	....	....	—1/3	1/3	1	5/3	—1	....	—2/3	....	—1/3	....	....	667
A_M ←	1	....	....	—2/3	....	....	....	....	....	—1/2	....	1/2	....	....	—1/2	....	—1/6	1/6	....	....
C_M	....	....	....	....	1	....	....	....	....	—1/2	....	—3/2	1	....	1/2	....	1/2	—1/2	....	....
X_N	....	....	....	....	....	....	....	....	—4/3	—2/3	....	2/3	....	1	—2/3	....	—1/3	....	....	667
D_M	....	....	....	—1/3	....	....	1	....	....	....	....	1	....	....	....	....	—1/3	1/3	....	1000
X_Q	....	....	....	....	....	....	....	....	....	2	....	6	—3	....	—4	1	—2	2	....	2000
D_N	....	....	....	1/3	....	....	....	1	....	....	....	....	....	....	....	....	1/3	—1/3	....	....
C_N	....	....	....	....	....	1	....	....	1/3	1/6	....	—1/6	....	....	1/6	....	—1/6	1/2	....	333
Z	....	....	....	....	....	....	....	....	10/3	5/3	....	—5/3	2	....	8/3	....	4/3	....	1	7333

Tableau IX

	A_M	A_N	B_M	B_N ↓	C_M	C_N	D_M	D_N	X_A	X_B	X_C	X_D	X_M	X_N	X_P	X_Q	X_R	X_S	Z	Q
A_N	1	1	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	1000
B_M	....	....	1	1	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	1000
X_C	—10/3	....	....	20/9	....	....	....	....	—1/3	2	1	....	—1	....	1	....	2/9	—5/9	....	667
X_D	2	....	....	—4/3	....	....	....	....	....	—1	....	1	....	....	—1	....	—1/3	1/3	....	....
C_M	3	....	....	—2	1	....	....	....	....	—2	....	....	1	....	—1	....	....	....	....	....
X_N	—4/3	....	....	8/9	....	....	....	....	—4/3	....	....	....	....	1	....	....	—1/9	—2/9	....	667
D_M	—2	....	....	1	....	....	1	....	....	1	....	....	....	....	1	....	....	....	....	1000
X_Q	—12	....	....	8	....	....	....	....	....	8	....	....	—3	....	2	1	....	....	....	2000
D_N ←	....	....	....	1/3	....	....	....	1	....	....	....	....	....	....	....	....	1/3	—1/3	....	....
C_N	1/3	....	....	—2/9	....	1	....	....	1/3	....	....	....	....	....	....	....	—2/9	5/9	....	333
Z	10/3	....	....	—20/9	....	....	....	....	10/3	....	....	....	2	....	1	....	7/9	5/9	1	7333

Tableau X

	A_M	A_N	B_M	B_N	C_M	C_N	D_M	D_N	X_A	X_B	X_C	X_D	X_M	X_N	X_P	X_Q	X_R	X_S ↓	Z	Q
A_N	1	1	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	1000
B_M	....	....	1	....	....	....	....	—3	....	1	....	....	....	....	....	....	—1	1	....	1000
X_C	—10/3	....	....	....	....	....	....	—20/3	—1/3	2	1	....	—1	....	1	....	—2	5/3	....	667
X_D	2	....	....	....	....	....	....	4	....	—1	....	1	....	....	—1	....	1	—1	....	....
C_M	3	....	....	....	1	....	....	6	....	—2	....	....	1	....	—1	....	2	—2	....	....
X_N	—4/3	....	....	....	....	....	....	—8/3	—4/3	....	....	....	....	1	....	....	—1	—2/3	....	667
D_M	—2	....	....	....	....	....	1	—3	....	1	....	....	....	....	1	....	—1	1	....	1000
X_Q ←	—12	....	....	....	....	....	....	—24	....	8	....	....	—3	....	2	1	—8	8	....	2000
B_N	....	....	....	1	....	....	....	3	....	....	....	....	....	....	....	....	1	—1	....	....
C_N	1/3	....	....	....	....	1	....	2/3	1/3	....	....	....	....	....	....	....	....	1/3	....	333
Z	10/3	....	....	....	....	....	....	20/3	10/3	....	....	....	2	....	1	....	3	—5/3	1	7333

Maximum Profit Feasible Solution

Tableau XI

	A_M	A_N	B_M	B_N	C_M	C_N	D_M	D_N	X_A	X_B	X_C	X_D	X_M	X_N	X_P	X_Q	X_R	X_S	Z	Q
A_N	1	1	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	1000
B_M	3/2	....	1	....	....	....	....	....	....	....	....	....	3/8	....	—1/4	—1/8	....	....	....	750
X_C	5/6	....	....	....	....	....	....	—5/3	—1/3	1/3	1	....	—3/8	....	7/12	—5/24	—1/3	....	....	250
X_D	1/2	....	....	....	....	....	....	1	....	....	....	1	—3/8	....	—3/4	1/8	....	....	....	250
C_M	....	....	....	....	1	....	....	....	....	....	....	....	1/4	....	—1/2	1/4	....	....	....	500
X_N	—1/3	....	....	....	....	....	....	—2/3	—4/3	—2/3	....	....	1/4	1	—1/6	—1/12	—1/3	....	....	500
D_M	—1/2	....	....	....	....	....	1	....	....	....	....	....	3/8	....	3/4	—1/8	....	....	....	750
X_S	—3/2	....	....	....	....	....	....	—3	....	1	....	....	—3/8	....	1/4	1/8	—1	1	....	250
B_N	—3/2	....	....	1	....	....	....	....	....	1	....	....	—3/8	....	1/4	1/8	....	....	....	250
C_N	5/6	....	....	....	....	1	....	5/3	1/3	—1/3	....	....	1/8	....	—1/12	—1/24	1/3	....	....	250
Z	5/6	....	....	....	....	....	....	5/3	10/3	5/3	....	....	11/8	....	17/12	5/24	4/3	....	1	7750

CHAPTER 5

A Refinery Improvement Program Under Competitive Conditions

For illustrative purposes this chapter presents the solution of a refinery problem in operations research by the use of the game theory solved by linear programming. The relationship between game theory and linear programming is discussed by Gale, Kuhn, and Tucker[8] and by George B. Dantzig.[9] One of the most appropriate reference books is *Introduction to the Theory of Games* by J. C. C. McKinsey.[11]

A refinery, X, produces insufficient yields of gasoline and distillate fuel to meet its requirements, while producing distress residual fuel. Refinery X must pay 10 cents per barrel excess transportation costs on products requiring shipment into or out of the marketing area in order to meet product demands. Four different proposals have been prepared for changing product distribution, which would result in increasing profit from 30 cents per barrel to 35, 39, 42, and 44 cents per barrel, respectively, including reductions in excess transportation costs from 10 cents per barrel to 6, 3, 1, and 0 cents per barrel respectively. Each proposal is economically justified if optimum operation is maintained. It seems obvious that the proposal showing the maximum profit should be adopted. However, the assumption made in this study was that a major competitor, refinery Y, who also has poor product distribution, would do nothing about improving its operation. It was possible to show that if refinery Y carried out the same type of improvement projects a reduction in excess transportation costs to refinery X, also, would result. This condition is due to the fact that the two refineries X and Y are sharing a fixed market demand, and any improvement in product yields at either refinery would decrease the excess transportation costs for both refineries. On the other hand, if both refineries carried out the maximum improvement program, the maldistribution of products would be overcorrected. Their combined products would again be out of balance, and both refineries would again incur excess transportation costs. Thus the excess transportation costs of refinery X depend upon the

strategies adopted by both refinery X and refinery Y. In this situation a profit matrix can be set up for five alternate strategies of X vs. the same five alternate strategies of Y. The problem of determining the best strategy for X can be solved by linear programming using the simplex method applied to a game theory problem. This method is described by Robert Dorfman in Koopmans, *Activity Analysis of Production and Allocation.*[3] The solution to the problem will also consider the possibilities of "mixed strategies," i.e., X may process part of its crude in one manner and part in another. The solution will show which methods will be best for refinery Y after X has adopted its best procedure and what the resulting profit for refinery X will be. If the profit matrix is correct, the answer will be certain to provide the best profit regardless of the competitor's intentions.

Given

Profit in cents per barrel at refinery X for each of five alternate strategies (no change and four proposals) depends upon the same five strategies at refinery Y. The following table shows the excess transportation costs.

	X_1	X_2	X_3	X_4	X_5
Y_1	10	6	3	1	0
Y_2	6	3	1	0	2
Y_3	3	1	0	2	5
Y_4	1	0	2	5	9
Y_5	0	2	5	9	14

In addition to these excess transportation costs, refinery X will increase its profit by 0, 1, 2, 3, and 4 cents per barrel by adopting strategies X_1, X_2, X_3, X_4, and X_5, respectively. The net effect of these changes on the current profit of 30 cents per barrel is shown in a profit table as follows:

	PROFIT* TO X				
	X_1	X_2	X_3	X_4	X_5
Y_1	30	35	39	42	44
Y_2	34	38	41	43	42
Y_3	37	40	42	41	39
Y_4	39	41	40	38	35
Y_5	40	39	37	34	30

*In cents per barrel.

Required

Determine the best strategies for X to maximize profit. Also determine the best responding strategies for Y and the resulting profit for X.

Development of the Mathematical Model

The profit for X will depend upon the strategy adopted by Y. Thus for any given y_i:

$$Z_i = \Sigma_j a_{ij} x_j \quad (x_j \geqq 0; \Sigma x_j = 1),$$

and for all values of y_i:

$$Z = \Sigma_i y_i \Sigma_j a_{ij} x_j \quad (y_i \geqq 0; \Sigma y_i = 1)$$

where x_j and y_i are the corresponding fractions of the strategies of X and Y, respectively, and a_{ij} are the specific profits to X for these strategies. The objective for X, of course, is to maximize Z. It will be assumed in this study that the objective for Y is to minimize Z. This assumption is based upon the direct competition of X and Y to share a fixed market demand. Thus, Z is the profit to X which is maximized by X and minimized by Y. Therefore:

$$Z_i - Z = b_i$$

where b_i is the excess of profit to X, if any, for selected strategy y_i. The best strategies for Y resulting in minimum profit to X will be obtained when $b_i = 0$.

Combining equations gives:

$$\Sigma_j a_{ij} x_j - b_i = Z.$$

Dividing by Z gives

$$\Sigma_j a_{ij} x_j / Z - b_i / Z = 1.$$

Substituting new variables X_j for x_j/Z and Y_i for b_i/Z results in a set of linear equations:

$$\Sigma_j a_{ij} X_j - Y_i = 1$$

where the objective for X is

$$\Sigma_j X_j = 1/Z = \min \quad (Z = \max)$$

and the objective for Y is

$$Y_i = 0 \qquad (b_i = 0).$$

Thus the desirable strategies for X will be in the basis and will have values

$$x_j = Z X_j,$$

while the desirable strategies for Y occur when $Y_i = 0$ and therefore will not be in the basis. They will have values

$$y_i = -Z \Sigma_j (-\triangle X_j / \triangle Y_i)$$

where $(-\triangle X_j / \triangle Y_i)$ are coefficients for X in the basis and Y not in the basis. This expression is equivalent to

$$\left(\frac{y_i}{Z}\right) = -(-\triangle \Sigma X / \triangle Y_i)$$

where $\Sigma X = \min$

or $$y_i = -\triangle Z / \triangle b_i$$

where $Z = \max$.

This relationship is obtained from the duality theorem of linear programming as discussed by Charnes and Cooper[2] and by Dantzig and Dorfman[3] where, at the solution of the game theoretic problem for all strategies in the basis,

$$\Sigma_j a_{ij} x_j^* = \Sigma_i a_{ij} y_i^* = Z.^*$$

Solution of the Problem

Construct a payoff matrix by subtracting a number (29) from each element in the profit table to reduce the minimum element to 1.

	X_1	X_2	X_3	X_4	X_5
Y_1	1	6	10	13	15
Y_2	5	9	12	14	13
Y_3	8	11	13	12	10
Y_4	10	12	11	9	6
Y_5	11	10	8	5	1

Construct a problem matrix by attaching $X_0 = 1$ and negative Y identity columns to the payoff matrix.

	X_0	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3	Y_4	Y_5
(Y_1)	1	1	6	10	13	15	−1	0	0	0	0
(Y_2)	1	5	9	12	14	13	0	−1	0	0	0
(Y_3)	1	8	11	13	12	10	0	0	−1	0	0
(Y_4)	1	10	12	11	9	6	0	0	0	−1	0
(Y_5)	1	11	10	8	5	1	0	0	0	0	−1

The problem matrix does not constitute a feasible solution because of the absence of identity vectors. Instead, there is a set of negative unit vectors forming a negative unit submatrix. In order to obtain a basic feasible solution it is necessary to convert the matrix to the usual form containing a unit submatrix. This can be done when making the first transformation to include an X in the basis. The first X to enter the basis should be selected with the largest minimum payoff. Thus if X selects strategy X_3, Y will select strategy Y_5 and the payoff to X will be 8. This is the best apparent single strategy. It will be noted that the Y_i to be excluded from the basis is determined by the *maximum* X_0/X_k in order to maintain all $X_0 \geqq 0$. The transformation is completed by use of the simplex algorithm for a'_{kj} and the *negative* simplex algorithm for all other a'_{ij}. Thus

$$a'_{kj} = \frac{a_{rj}}{a_{rk}}$$

and

$$a'_{ij} = \frac{a_{rj}}{a_{rk}} a_{ik} - a_{ij} \quad (i \neq k).$$

These algorithms will definitely convert the Y vectors to unit vectors which together with the substituted X vector will form a unit submatrix and thus provide a basis for the feasible solution.

The basic feasible solution is expressed in the following matrix form.

	X_0	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3	Y_4	Y_5
X_3	0.125	1.375	1.25	1	0.625	0.125	0	0	0	0	−0.125
Y_1	0.250	12.75	6.50	0	−6.75	−13.75	1	0	0	0	−1.25
Y_2	0.500	11.50	6.00	0	−6.50	−11.50	0	1	0	0	−1.50
Y_3	0.625	9.875	5.25	0	−3.875	−8.375	0	0	1	0	−1.625
Y_4	0.375	5.125	1.75	0	−2.125	−4.625	0	0	0	1	−1.375
ΣX	0.125	1.375	1.25	1	0.625	0.125	0	0	0	0	−0.125

The row ΣX is the same as row X_3 since only a single strategy is involved in this solution. The value of the game to X by adopting strategy X_3 is $1/\Sigma X$ in the X_0 column or $1/0.125 = 8.0$. The profit to X is $29 + 8.0 = 37.0$ cents per barrel. The value of the game to Y by adopting responding strategy Y_5 is $1/\Sigma X$ in the Y columns or $1/-0.125 = -8.0$. (The profit to X is considered a loss to Y.)

In order to be a maximum profit solution, ΣX must be not higher than 1 in any of the X columns and not higher than 0 in any of the Y columns. All of the ΣX elements meet these requirements except in the X_1 and X_2 columns where ΣX is 1.375

and 1.25, respectively. Strategy X_1 is chosen, therefore, to enter the next feasible solution (No. 2). The row to omit from the new matrix is determined as usual by the smallest ratio of its element in the X_0 column to its element in the X_1 (new strategy) column. The smallest ratio is in the Y_1 row (0.250/12.75 = 0.0196) and the Y_1 row is therefore omitted from the new matrix. This means that Y_1 becomes one of Y's responding strategies in feasible solution No. 2.

The matrix for feasible solution No. 2 may be completed by use of the regular simplex algorithms:

	X_0	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3	Y_4	Y_5	ΣY
X_1	0.0196	1	0.510	0	−0.529	−1.078	0.0784	0	0	0	−0.0980	
X_3	0.0980	0	0.549	1	1.353	1.608	−0.1078	0	0	0	0.0098	
Y_2	0.275	0	0.135	0	−0.412	0.902	−0.902	1	0	0	−0.373	
Y_3	0.431	0	0.214	0	1.353	2.275	−0.775	0	1	0	−0.657	
Y_4	0.275	0	−0.864	0	0.588	0.902	−0.402	0	0	1	−0.873	
ΣX	0.1176	1	1.059	1	0.824	0.530	−0.294	0	0	0	−0.0882	−0.1176

In this solution X uses strategy X_1 for (0.0196/0.1176)100 = 16.7 per cent of its crude and strategy X_3 for (0.0980/0.1176)100 = 83.3 per cent of its crude. Y uses responding strategies Y_1 for (−0.0294/−0.1176)100 = 25.0 per cent of its crude and Y_5 for (−0.0882/−0.1176)100 = 75.0 per cent of its crude. The value of the game to X is 1/0.1176 = 8.50, corresponding to a profit of 29+ 8.50 = 37.50 cents per barrel. However, this solution is not a maximum profit feasible solution since ΣX in the X_2 column is greater than 1, indicating that the X_2 strategy should be used in a new feasible solution. Since the ratio of X_0 to X_2 in the X_1 row of 0.0196/0.510 = 0.0385 is the smallest ratio, strategy X_1 should be omitted from the new feasible solution.

The matrix for feasible solution No. 3 may be completed.

	X_0	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3	Y_4	Y_5	ΣY
X_2	0.0385	1.961	1	0	−0.137	−2.114	0.1535	0	0	0	−0.1920	
X_3	0.0770	−1.076	0	1	1.923	2.768	−0.1920	0	0	0	0.1150	
Y_2	0.269	−0.265	0	0	−0.272	1.187	−0.923	1	0	0	−0.347	
Y_3	0.423	−0.419	0	0	1.575	2.726	−0.807	0	1	0	−0.616	
Y_4	0.308	1.686	0	0	−0.307	−0.923	−0.270	0	0	1	−1.039	
ΣX	0.1155	0.885	1	1	0.886	0.654	−0.0385	0	0	0	−0.0770	−0.1155

Since ΣX is not more than 1 in any X column and not more than 0 in any Y column, solution No. 3 becomes a maximum profit feasible solution. The optimum strategies for X are X_2 for (0.0385/0.1155)100 = 33.3 per cent of its crude and X_3 for (0.7700/0.1155)100 = 66.7 per cent of its crude. The optimum responding strategies for Y are Y_1 for (—0.0385/—0.1155)100 = 33.3 per cent of its crude and Y_5 for (—0.0770/—0.1155)100 = 66.7 per cent of its crude. The value of the game to X will be 1/0.1155 = 8.66 corresponding to a profit of 29 + 8.66 = 37.66 cents per barrel which is the maximum profit possible with the profit table given in this problem.

This problem can also be solved with refinery Y as the maximizing player. It is necessary to restate the problem matrix to include a payoff matrix to Y recoded so that the largest payoff to X will be —1 (equivalent to +1 to Y). By subtracting 45 from the original profit table and transposing the negative matrix, the following payoff matrix to Y is obtained.

	Y_1	Y_2	Y_3	Y_4	Y_5
X_1	15	11	8	6	5
X_2	10	7	5	4	6
X_3	6	4	3	5	8
X_4	3	2	4	7	11
X_5	1	3	6	10	15

In the first transformation Y_5 should replace X_1. The optimal solution, of course, is identical with that of the problem maximizing the profit to refinery X.

In the event that the information in the profit table may not be completely known by both competitors, there may be considerable advantage in computing responding (defensive) strategies based upon an estimate of competitor's knowledge of the payoff matrix. Since the offensive strategies are always based upon the best strategies for the defensive, any defect in knowledge of the offensive side will result in extra profit for the defensive side.

A Two-Player Non-Zero-Sum Game Theory Problem

In many cases, the two-player zero-sum game theoretic method cannot be used for two sets of strategies because, though the strategies are related, the profit matrices for the two players are not opposite. Such a situation is illustrated below for two business competitors who have separate alternate but interrelated plans for improving the profit from their respective businesses.

	PROFIT* TO X					PROFIT* TO Y			
	x_1	x_2	x_3	x_4		y_1	y_2	y_3	y_4
y_1	1	2	3	5	x_1	4	3	2	1
y_2	2	1	2	3	x_2	3	2	1	2
y_3	3	2	1	2	x_3	2	1	2	4
y_4	4	3	2	1	x_4	1	2	4	7

*In thousands of dollars per day.

Each problem is set up for solution by linear programming, giving the following solution matrices.

For the *X* game

	x_1	x_2	x_3	x_4	b_1	b_2	b_3	b_4	b_5	Z_x	Q
b_1	....	−2	1	....	1	−5/2	3/2	....	1/2	....	1/2
Z_x	....	1	1	....	....	1/2	1/2	....	5/2	1	5/2
x_4	....	....	1	1	....	−1/2	1/2	....	1/2	....	1/2
b_4	....	....	−2	....	....	1	−2	1	....	....	0
x_1	1	1	....	....	....	1/2	−1/2	....	1/2	....	1/2

and for the *Y* game

	y_1	y_2	y_3	y_4	b_1	b_2	b_3	b_4	b_5	Z_y	Q
y_4	....	....	2/3	1	....	1/3	−1/3	....	1/3	....	1/3
b_4	....	−2	−1/3	....	....	4/3	−7/3	1	1/3	....	1/3
y_1	1	1	1/3	....	....	−1/3	1/3	....	2/3	....	2/3
Z_y	....	1	4/3	....	....	2/3	1/3	....	8/3	1	8/3
b_1	....	....	−4/3	....	1	−5/3	2/3	....	1/3	....	1/3

The minimax profit* for X is 5/2 ($2500 per day) for strategies $x_1 = 1/2$ and $x_4 = 1/2$, and the minimax profit for Y is 8/3 ($2667 per day) for strategies $y_1 = 2/3$ and $y_4 = 1/3$. It will be noted that X would lose 1 ($1000 per day) for the adoption of either x_2 or x_3 strategies and that Y would lose 1 ($1000 per day) for the adoption of y_2 strategy or 4/3 ($1333 per day) for the y_3 strategy. It may safely be assumed, therefore, that X will adopt x_1 and x_4 strategies and Y will adopt y_1 and y_4 strategies.

In view of these results, it is advisable to set up reduced games for the x_1, x_4, y_1, and y_4 strategies having the following profit matrices.

*That is, the maximum of the alternate minimum values.

STRATEGIES OF X		
	x_1	x_4
y_1	1	5
y_4	4	1

STRATEGIES OF Y		
	y_1	y_4
x_1	4	1
x_4	1	7

The solution of these games results in the following solution matrix for X (reduced game):

	x_1	x_4	b_1	b_4	b_5	Z_x	Q
Z_x	....	....	3/7	4/7	19/7	1	19/7
x_4	....	1	−1/7	1/7	3/7	....	3/7
x_1	1	....	1/7	−1/7	4/7	....	4/7

and for Y (reduced game)

	y_1	y_4	b_1	b_4	b_5	Z_y	Q
y_4	....	1	1/9	−1/9	1/3	....	1/3
Z_y	....	....	2/3	1/3	3	1	3
y_1	1	....	−1/9	1/9	2/3	....	2/3

As a result of restricting the responding strategies, the profit for X has increased from 5/2 to 19/7 ($2714 per day) and the profit for Y has increased from 8/3 to 3 ($3000 per day). When using the values of $x_1 = 4/7$, $x_4 = 3/7$, $y_1 = 2/3$, and $y_4 = 1/3$, the following minimax profits result:

PROFIT TO X		PROFIT TO Y	
x_1y_1	$4/7 \times 2/3 \times 1 = 8/21$	x_1y_1	$4/7 \times 2/3 \times 4 = 32/21$
x_4y_4	$3/7 \times 2/3 \times 5 = 30/21$	x_4y_1	$3/7 \times 2/3 \times 1 = 6/21$
x_1y_4	$4/7 \times 1/3 \times 4 = 16/21$	x_1y_4	$4/7 \times 1/3 \times 1 = 4/21$
x_4y_4	$3/7 \times 1/3 \times 1 = 3/21$	x_4y_4	$3/7 \times 1/3 \times 7 = 21/21$
	Total $Z_x = 19/7$		Total $Z_y = 3$

It is desired to select strategies which will assure at least the minimax profits to X and Y. In fact, when using the strategies indicated in the solution matrices for the reduced games, no other profits can result. This is shown as follows:

Profit to X			Profit to Y		
	$y_1 = 1$ or	$y_4 = 1$		$x_1 = 1$ or	$x_4 = 1$
$x_1 = 4/7$	$4/7 \times 1$	$4/7 \times 4$	$y_1 = 2/3$	$2/3 \times 4$	$2/3 \times 1$
$x_4 = 3/7$	$3/7 \times 5$	$3/7 \times 1$	$y_4 = 1/3$	$1/3 \times 1$	$1/3 \times 7$
Total Z_x	19/7 or	19/7	Total Z_y	3 or	3

However, if X uses the responding strategies in the Y reduced game or if Y uses the responding strategies in the X reduced game, additional profits for either X or Y can be obtained. The possible increased profit for X by varying Y strategies can be shown as follows:

	Profit to X			**Profit to Y**			
	$y_1 = 1$ or	$y_4 = 1$			$x_1 = 2/3$	$x_4 = 1/3$	Total Z_y
$x_1 = 2/3$	$2/3 \times 1$	$2/3 \times 4$		$y_1 = 1$	$2/3 \times 4$	$1/3 \times 1$	3
				or			or
$x_4 = 1/3$	$1/3 \times 5$	$1/3 \times 1$		$y_4 = 1$	$2/3 \times 1$	$1/3 \times 7$	3
Total Z_x	7/3 or	3					

Thus if X adopts the responding strategies $x_1 = 2/3$ and $x_4 = 1/3$, player Y can then vary his strategies without loss to Y. If Y can be induced under this situation to adopt strategies $y_1 = 0$ and $y_4 = 1$, then the profit for X will be increased from 19/7 to 3 ($3000 per day).

Similarly, the possible increased profit for Y by varying X strategies can be shown as follows:

	Profit to X				**Profit to Y**	
	$y_1 = 3/7$	$y_4 = 4/7$	Total Z_x		$x_1 = 1$ or	$x_4 = 1$
$x_1 = 1$	$3/7 \times 1$	$4/7 \times 4$	19/7	$y_1 = 3/7$	$3/7 \times 4$	$3/7 \times 1$
or			or			
$x_4 = 1$	$3/7 \times 5$	$4/7 \times 1$	19/7	$y_4 = 4/7$	$4/7 \times 1$	$4/7 \times 7$
				Total Z_y	16/7 or	31/7

Thus if Y adopts the responding strategies $y_1 = 3/7$ and $y_4 = 4/7$, player X can then vary his strategies without loss to X. If X can be induced under this situation to adopt strategies $x_1 = 0$ and $x_4 = 1$, then the profit for Y will be increased from 3 to 31/7 ($4429 per day).

An incentive can be provided by either X or Y changing his responding strategies slightly so that his opponent will select a strategy resulting in maximum profit. Thus, by changing X's responding strategies by 1/21 the situation outlined at the top of page 35 results.

In the case below, it is desired, of course, to induce Y to adopt strategies $y_1 = 0$ and $y_4 = 1$. Incentive is provided for this move when X increases strategy x_4 slightly above the responding value of 1/3. By increasing x_4 to 8/21 and decreasing x_1 to 13/21, there is a distinct profit incentive in the value of 69/21 ($3286 per day) for

	Profit to X			Profit to Y		
	$y_1 = 1$ or $y_4 = 1$			$x_1 = 13/21$	$x_4 = 8/21$	total Z_y
$x_1 = 13/21$	$13/21 \times 1$	$13/21 \times 4$	$y_1 = 1$ or	$13/21 \times 4$	$8/21 \times 1$	$60/21$ or
$x_4 = 8/21$	$8/21 \times 5$	$8/21 \times 1$	$y_4 = 1$	$13/21 \times 1$	$8/21 \times 7$	$69/21$
Total Z_x	$53/21$ or	$20/7$				

Y to adopt strategy $y_4 = 1$. This results in a profit to X of 20/7 ($2857 per day) compared with a minimax value of 19/7 ($2714 per day) and a maximum value (without incentive or loss to Y) of 3 ($3000 per day).

Incentive for X to adopt a strategy most favorable to Y can be provided if Y should change his responding strategies by 1/21 resulting in the following situation:

	Profit to X				Profit to Y	
	$y_1 = 10/21$	$y_4 = 11/21$	Total Z_x		$x_1 = 1$ or	$x_4 = 1$
$x_1 = 1$ or	$10/21 \times 1$	$11/21 \times 4$	$18/7$ or	$y_1 = 10/21$	$10/21 \times 4$	$10/21 \times 1$
$x_4 = 1$	$10/21 \times 5$	$11/21 \times 1$	$61/21$	$y_2 = 11/21$	$11/21 \times 1$	$11/21 \times 7$
				Total Z_y	$17/7$ or	$29/7$

In this case Y has increased strategy y_1 by 1/21 from 3/7 to 10/21, thus causing the profit to X to increase from 19/7 ($2714 per day) to 61/21 ($2905 per day) and the resulting profit to Y to increase to 29/7 ($4143 per day) compared with a minimax value of 3 ($3000 per day) and a maximum value (without incentive or loss to X) of 31/7 ($4429 per day).

A summary of these various strategies is shown on page 36.

It can be seen that, if both X and Y adopt responding strategies, the profit to each is the same as the minimax profit for the reduced games without the safeguard provided by the minimax strategies. Either player could suffer loss in his profit if his opponent altered his strategy to bring this about. If both players alter their responding strategies slightly to induce their opponents to select more favorable strategies, the result would be only a slight change from the minimax profits. In this example it is obvious that Y can offer more inducement to X than X can offer to Y so that the final best arrangement of strategies would be $X_1 = 0$ and $X_4 = 1$ for X and approximately $Y_1 = 10/21$ and $Y_4 = 11/21$ for Y to provide profits considerably over the minimax profits for both X and Y. It is possible, of course, to select single strategies for the players which will give more profit to one player or to both players

SUMMARY OF VARIOUS STRATEGIES

	x_1	x_4	PROFIT TO X*	y_1	y_4	PROFIT TO Y*
FULL GAMES						
Minimax for X	1/2	1/2	2500	would play y_2 and y_3		
Minimax for Y	would play x_2 and x_3			2/3	1/3	2667
REDUCED GAMES						
Minimax for X	4/7	3/7	2714	1 or 1		
Minimax for Y	1 or 1			2/3	1/3	3000
USING RESPONDING STRATEGIES						
Minimum to X	2/3	1/3	2333	1	0	3000
Maximum to X	2/3	1/3	3000	0	1	3000
With inducement to Y	13/21	8/21	2857	0	1	3286
Minimum to Y	1	0	2714	3/7	4/7	2286
Maximum to Y	0	1	2714	3/7	4/7	4429
With inducement to X	0	1	2905	10/21	11/21	4143
Both X and Y responding	2/3	1/3	2714	3/7	4/7	3000
WITH COLLUSION						
Maximum profit to X	0	1	5000	1	0	1000
Maximum profit to Y	0	1	1000	0	1	7000

*In dollars per day.

together, but this will require collusion since both individual interests cannot be served simultaneously by adopting single strategies.

With the information obtained from the game theoretic analysis of a competitive situation as outlined here, decisions can more easily be made to adopt strategies that will assure minimax profits regardless of the competitor's actions, or to adopt strategies that will take advantage of any increased profits made available by competitor's selection of non-minimax strategies.

CHAPTER 6

Optimum Production Rates and Inventory to Meet Uncertain Seasonal Requirements

The selection of production rates and inventory to meet a variable seasonal requirement for a product such as heating fuel may have a major effect upon the profit or loss of the operation. In general, variations in the production rate are uneconomical and may be reduced by increasing inventory. Large inventories are also costly, however, because of the operating expense and interest on the marginal capital tied up in the storage facilities and in the stock.

The production cost curve for heating fuel passes through the minimum value at an average production rate which is equal to the average requirement. This condition has been brought about by expansion of the heating fuel market in relation to other products as long as the unit production cost could be decreased. The extra cost at low production rates is due to uneconomical processing methods to dispose of the surplus and the extra cost at high production rates is due to diversion of more valuable components to heating fuel.

For this illustrative study the heating season has been divided into two equal periods of 90 days each, denoted as period A and period B. The remainder of the year is denoted as period C. It is necessary to decide what is the maximum size of the inventory to be accumulated before period A, denoted as inventory 1, and the production rates to run during periods A, B, and C. The requirements for the product have been found to vary in almost random fashion above and below the average for each of periods A and B. No significant effect of requirement A has been found in requirement B. A more complete study of this problem would divide the heating season into many parts, and in this case it would be necessary to consider the persistence of requirements from one period to the next as well as the process lag in changing from one production rate to another. It is also possible that the inventory and production rate may have some specific effects upon the requirements.

The possibility of failing to meet requirements was considered; but because the method of evaluating political and good will effects was not known this case was not

included in the model. The variability of the requirements considered in the model covers about 99.7 per cent of the expected occurrences in requirements. The extreme cases not considered would not affect optimum operating practice. For all other cases production rates were allowed to vary only within limits of meeting possible expected requirements. The production rate during period B was never allowed to fall below the average rate during period C since with a low production rate necessary to meet requirement the inventory would not be critical and a more even production rate during periods B and C would be economical. In this case the average production rate could only be approximated until the requirement in period B had developed sufficiently.

The realization price of the heating fuel was assumed to be constant for period A and also for period B except where greatest deviation in requirement was found in period A. A 25 per cent deviation in requirement in period A was accompanied by a 5 per cent change in price in period B. These cases constitute no more than 5 per cent of the occurrences in either direction. In a more complete study the possible effect of inventory and production rate on the price-requirement relationship would be investigated.

Development of the Mathematical Model

It was decided to formulate this problem as a constrained game for solution by linear programming. (Professor A. Charnes has excellent notes for several lectures on constrained games[12].) This method will allow for incorporation of additional factors in the complete problem. After some preliminary study, six proposed alternate strategies were included for evaluation, involving two different sets of production rates at each of three different maximum inventories. Thirteen different probable occurrences of requirements for periods A and B were included, with restrictions placed on these occurrences in accordance with the histogram shown on Chart I, page 39. The payoff matrix was built up from the production rate and inventory cost and price data shown on Chart II, page 39. The linear equations were developed as follows:

The profit to the refiner is Z. In order to allow for negative profit (loss to refiner)

$$Z = x_{n+1} - x_{n+2}$$

where the non-negative variables x_{n+1} and x_{n+2} are profit and loss, respectively.

The constraints in this problem include the limiting probabilities on nature's strategies

$$\Sigma\, x_j \leqq p_i \qquad \text{(for certain } x_j\text{)}$$

and the payoff matrix

$$\Sigma_j a_{ij} x_j + b_i = Z.$$

Since nature is the maximizing player in this game, the objective will be to maximize

CHART I

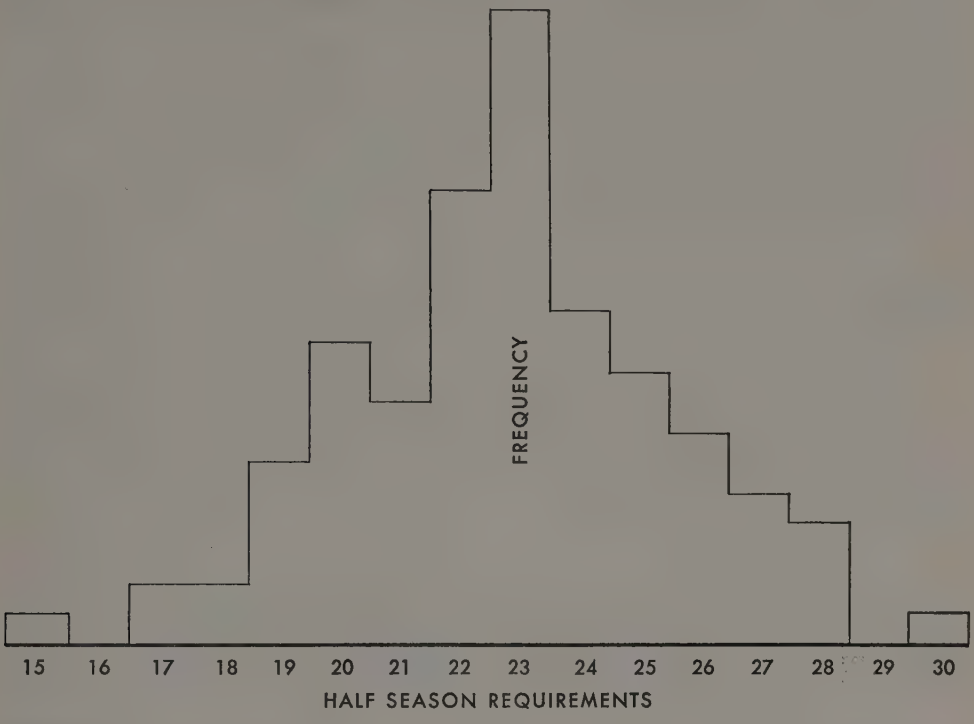
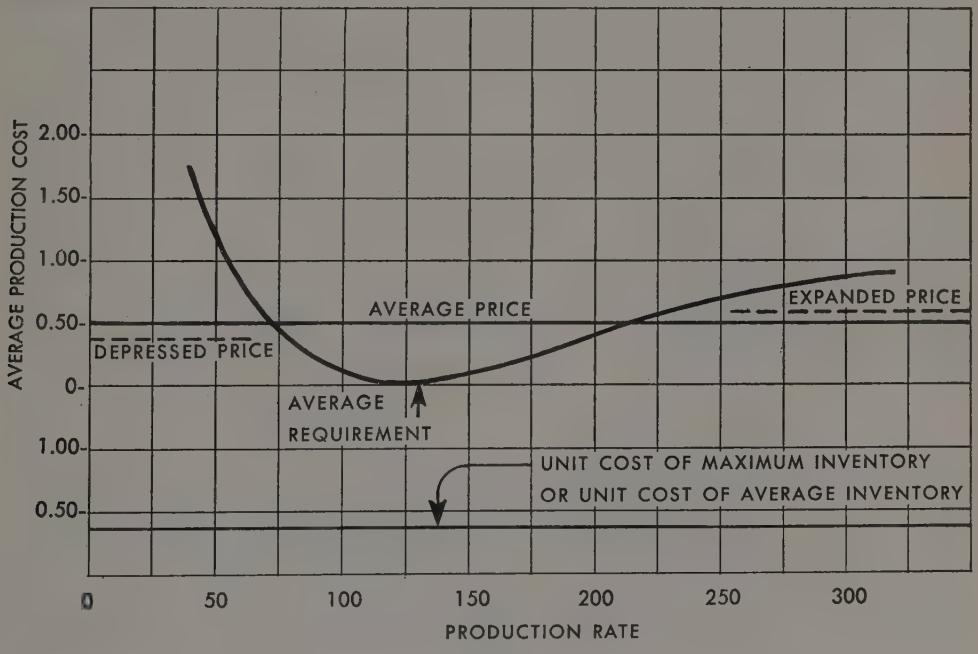


CHART II



$(-Z)$. In addition, an artificial vector† with a unit profit of $-M$ will be used to force $\Sigma x_j = 1$. Thus

$$\Sigma x_j + w = 1 \quad (w \to 0).$$

The profit functional to nature therefore becomes

$$f(-Z) = -x_{n+1} + x_{n+2} - Mw = \max.$$

In the basic problem matrix, this functional has the value

$$f(-Z) = -M$$

since $w=1$ in the null case. The equation for the $f(-Z)$ row in the problem matrix therefore becomes

$$-M = f(-Z) - M\Sigma x_j + x_{n+1} - x_{n+2}.$$

The unit vector for $f(-Z)$ which will not be excluded from the basis has been omitted from the attached problem and solution matrices but is implied. The variable w will be excluded from the basis in the optimal solution because of its $-M$ unit profit. This gives an optimal value

$$f^*(-Z) = -Z^*.$$

The optimal refinery strategy will be obtained when $b_i = 0$. Positive b_i values indicate the decrease from profit Z for the other strategies.

The method of building up the payoff matrix can be illustrated as follows for Refinery Strategy y_{14} and Nature's Strategy x_{20}

		Quantity	Cost (above normal)
y_{14}	Max inventory1	24	8.64
	Production A	10.5	0.13
x_{20}	Requirement A	16.5	
	Requirement B	22.5	
	Inventory 2	18	
	Possible requirement	28.5	
	Production B	10.5	0.13
	Inventory 3	6	
	Production C	18	1.67
	Avg. inventory	15	5.40
	Total production	39	15.97
	Normal realization	39	19.50
	Price depression	22.5	−4.50
	Profit		−0.97

†See Charnes and Cooper, *Introduction to Linear Programming*[2] for a discussion of the artificial vector.

The complete calculations for the payoff matrix are shown on Tables I and II.

The restrictions on nature's 13 strategies were derived from a study of past requirements. It was estimated that no more than 25 per cent of the occurrences of the half-season requirements will be significantly below the normal of 22.5 and no more than 25 per cent will be significantly above the normal. Most of these deviations will average about ± 3 from the normal and not more than 20 per cent of them will average about ± 6. The linear equations for these restrictions were developed as follows:

Probability of half-season deviation of 3 or more	$p_D \leq 0.25$
Probability of half-season deviation of 6 or more	$p_{2D} \leq 0.05$
Probability of half-season deviation of 3 plus half-season deviation of 3 or more	$(p_D - p_{2D})\,(p_D) \leq 0.05$
Probability of normal half-season plus half-season deviation of 3 or more	$(1.0 - 2p_D)\,(p_D) \leq 0.125$
Probability of normal half-season plus half-season deviation of 6 or more	$(1.0 - 2p_D)\,(p_{2D}) \leq 0.025$

Nature's strategies, x_{20} to x_{32} shown in the problem matrix, Table III, include probable occurrences in various combinations of requirements A and B restricted by the above inequalities.

Solution of the Problem

The solution of this problem was carried out by the IBM 701 Computer. The matrix for the optimal feasible solution is shown in Table IV. A summary of this solution follows:

Refinery Strategy	Inventory 1	Production A	Probable Profit
y_{16}	27	10.5	$3.786 - 0 = 3.786$
y_{17}	27	13.5	$3.786 - 0.164 = 3.622$
y_{18}	30	10.5	$3.786 - 0.252 = 3.534$
y_{15}	24	13.5	$3.786 - 0.965 = 2.821$
y_{19}	30	13.5	$3.786 - 1.822 = 1.964$
y_{14}	24	10.5	$3.786 - 2.919 = 0.867$

The best of the refinery strategies included in this study is obviously y_{16} with a probable profit of 3.786. The effect of varying inventory 1 and production A can be determined by comparing the cases. Thus increasing production A from 10.5 to 13.5 with inventory 1 at 27 would decrease probable profit by 0.164. However, with inventory 1 at 24 the optimum production A would be 13.5, and the decrease in prob-

TABLE I: Inventories and Production Rates Needed to Meet Various Requirements

Refinery Strategy			Nature's Strategy	P_{20}	P_{21}	P_{22}	P_{23}	P_{24}	P_{25}	P_{26}	P_{27}	P_{28}	P_{29}	P_{30}	P_{31}	P_{32}
			Requirement A	16.5	19.5	19.5	19.5	22.5	22.5	22.5	22.5	22.5	25.5	25.5	25.5	28.5
			Requirement B	22.5	19.5	22.5	25.5	16.5	19.5	22.5	25.5	28.5	19.5	22.5	25.5	22.5
	Inv. 1	Prod. A														
P_{14}	24	10.5	Inv. 2	18	15	15	15	12	12	12	12	12	9	9	9	6
			Prod. B	10.5	13.5	13.5	13.5	16.5	16.5	16.5	16.5	16.5	19.5	19.5	19.5	22.5
			Inv. 3	6	9	6	3	12	9	6	3	0	9	6	3	6
			Prod. C	18	15	18	21	12	15	18	21	24	15	18	21	18
			Avg. Inv.	15	16.5	15	13.5	18	16.5	15	13.5	12	16.5	15	13.5	15
P_{15}	24	13.5	Inv. 2	21	18	18	18	15	15	15	15	15	12	12	12	9
			Prod. B	8.5	10.5	10.5	10.5	13.5	13.5	13.5	13.5	13.5	16.5	16.5	16.5	19.5
			Inv. 3	7	9	6	3	12	9	6	3	0	9	6	3	6
			Prod. C	17	15	18	21	12	15	18	21	24	15	18	21	18
			Avg. Inv.	15.5	16.5	15	13.5	18	16.5	15	13.5	12	16.5	15	13.5	15
P_{16}	27	10.5	Inv. 2	21	18	18	18	15	15	15	15	15	12	12	12	9
			Prod. B	9.5	10.5	10.5	11.5	13.5	13.5	13.5	13.5	13.5	16.5	16.5	16.5	19.5
			Inv. 3	8	9	6	4	12	9	6	3	0	9	6	3	6
			Prod. C	19	18	21	23	15	18	21	24	27	18	21	24	21
			Avg. Inv.	17.5	18	16.5	15.5	19.5	18	16.5	15	13.5	18	16.5	15	16.5
P_{17}	27	13.5	Inv. 2	24	21	21	21	18	18	18	18	18	15	15	15	12
			Prod. B	8.5	8.5	9.5	10.5	10.5	10.5	10.5	11.5	12.5	13.5	13.5	13.5	16.5
			Inv. 3	10	10	8	6	12	9	6	4	2	9	6	3	6
			Prod.C	17	17	19	21	15	18	21	23	25	18	21	24	21
			Avg. Inv.	18.5	18.5	17.5	16.5	19.5	18	16.5	15.5	14.5	18	16.5	15	16.5
P_{18}	30	10.5	Inv. 2	24	21	21	21	18	18	18	18	18	15	15	15	12
			Prod. B	9.5	9.5	10.5	11.5	10.5	10.5	11.5	12.5	13.5	13.5	13.5	13.5	16.5
			Inv. 3	11	11	9	7	12	9	7	5	3	9	6	3	6
			Prod. C	19	19	21	23	18	21	23	25	27	21	24	27	24
			Avg. Inv.	20.5	20.5	19.5	18.5	21	19.5	18.5	17.5	16.5	19.5	18	16.5	18
P_{19}	30	13.5	Inv. 2	27	24	24	24	21	21	21	21	21	18	18	18	15
			Prod. B	8.5	8.5	9.5	10.5	8.5	9.5	10.5	11.5	12.5	10.5	11.5	12.5	13.5
			Inv. 3	13	13	11	9	13	11	9	7	5	9	7	5	6
			Prod. C	17	17	19	21	17	19	21	23	25	21	23	25	24
			Avg. Inv.	21.5	21.5	20.5	19.5	21.5	20.5	19.5	18.5	17.5	19.5	18.5	17.5	18

TABLE II: Cost and Profit Data for Inventories, Production Rates and Requirements

Refinery Strategy	Inv. 1	Prod. A	Nature's Strategy	P_{20}	P_{21}	P_{22}	P_{23}	P_{24}	P_{25}	P_{26}	P_{27}	P_{28}	P_{29}	P_{30}	P_{31}	P_{32}
			Requirement A	**16.5**	**19.5**	**19.5**	**19.5**	**22.5**	**22.5**	**22.5**	**22.5**	**22.5**	**25.5**	**25.5**	**25.5**	**28.5**
			Requirement B	**22.5**	**19.5**	**22.5**	**25.5**	**16.5**	**19.5**	**22.5**	**25.5**	**28.5**	**19.5**	**22.5**	**25.5**	**22.5**
P_{14}	24	10.5	Prod. A	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13
			Prod. B	0.13	1.02	1.02	1.02	4.30	4.30	4.30	4.30	4.30	9.44	9.44	9.44	14.58
			Prod. C	1.67	4.29	1.67	0.26	7.48	4.29	1.67	0.26	0.19	4.29	1.67	0.26	1.67
			Max. Inv.	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64
			Avg. Inv.	5.40	5.94	5.40	4.86	6.48	5.94	5.40	4.86	4.32	5.94	5.40	4.86	5.40
			Cost	15.97	20.02	16.86	14.91	27.03	23.30	20.14	18.19	17.58	28.44	25.28	23.33	30.42
			Price	15.00	19.50	21.00	22.50	19.50	21.00	22.50	24.00	25.50	22.50	24.00	25.50	30.00
			Profit	—0.97	—0.52	4.14	7.59	—7.53	—2.30	2.36	5.81	7.92	—5.94	—1.28	2.17	—0.42
P_{15}	24	13.5	Prod. A	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02
			Prod. B	0.86	0.13	0.13	0.13	1.02	1.02	1.02	1.02	1.02	4.30	4.30	4.30	9.44
			Prod. C	2.22	4.29	1.67	0.26	7.48	4.29	1.67	0.26	0.19	4.29	1.67	0.26	1.67
			Max. Inv.	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64	8.64
			Avg. Inv.	5.58	5.94	5.40	4.86	6.48	5.94	5.40	4.86	4.32	5.94	5.40	4.86	5.40
			Cost	18.32	20.02	16.86	14.91	24.64	20.91	17.75	15.80	15.19	24.19	21.03	19.08	26.17
			Price	15.00	19.50	21.00	22.50	19.50	21.00	22.50	24.00	25.50	22.50	24.00	25.50	30.00
			Profit	—3.32	—0.52	4.14	7.59	—5.14	0.09	4.75	8.20	10.31	—1.69	2.97	6.42	3.83
P_{16}	27	10.5	Prod. A	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13
			Prod. B	0.40	0.13	0.13	0.10	1.02	1.02	1.02	1.02	1.02	4.30	4.30	4.30	9.44
			Prod. C	1.12	1.67	0.26	0	4.29	1.67	0.26	0.19	1.59	1.67	0.26	0.19	0.26
			Max. Inv.	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72
			Avg. Inv.	6.30	6.48	5.94	5.58	7.02	6.48	5.94	5.40	4.86	6.48	5.94	5.40	5.94
			Cost	17.67	18.13	16.18	15.53	22.18	19.02	17.07	16.46	17.32	22.30	20.35	19.74	25.49
			Price	15.00	19.50	21.00	22.50	19.50	21.00	22.50	24.00	25.50	22.50	24.00	25.50	30.00
			Profit	—2.67	1.37	4.82	6.97	—2.68	1.98	5.43	7.54	8.18	0.20	3.65	5.76	4.51
P_{17}	27	13.5	Prod. A	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02
			Prod. B	0.86	0.86	0.40	0.13	0.13	0.13	0.13	0.10	0.42	1.02	1.02	1.02	4.30
			Prod. C	2.22	2.22	1.12	0.26	4.29	1.67	0.26	0	0.53	1.67	0.26	0.19	0.26
			Max. Inv.	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72	9.72
			Avg. Inv.	6.66	6.66	6.30	5.94	7.02	6.48	5.94	5.58	5.22	6.48	5.94	5.40	5.94
			Cost	20.48	20.48	18.56	17.07	22.18	19.02	17.07	16.42	16.91	19.91	17.96	17.35	21.24
			Price	15.00	19.50	21.00	22.50	19.50	21.00	22.50	24.00	25.50	22.50	24.00	25.50	30.00
			Profit	—5.48	—0.98	2.44	5.43	—2.68	1.98	5.43	7.58	8.59	2.59	6.04	8.15	8.76
P_{18}	30	10.5	Prod. A	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13
			Prod. B	0.40	0.40	0.13	0.10	0.13	0.13	0.10	0.42	1.02	1.02	1.02	1.02	4.30
			Prod. C	1.12	1.12	0.26	0	1.67	0.26	0	0.53	1.59	0.26	0.19	1.59	0.19
			Max. Inv.	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80
			Avg. Inv.	7.38	7.38	7.02	6.66	7.56	7.02	6.66	6.30	5.94	7.02	6.48	5.94	6.48
			Cost	19.83	19.83	18.34	17.69	19.29	18.34	17.69	18.18	19.48	19.23	18.62	19.48	21.90
			Price	15.00	19.50	21.00	22.50	19.50	21.00	22.50	24.00	25.50	22.50	24.00	25.50	30.00
			Profit	—4.83	—0.33	2.66	4.81	0.21	2.66	4.81	5.82	6.02	3.27	5.38	6.02	8.10
P_{19}	30	13.5	Prod. A	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02
			Prod. B	0.86	0.86	0.40	0.13	0.86	0.40	0.13	0.10	0.42	0.13	0.10	0.42	1.02
			Prod. C	2.22	2.22	1.12	0.26	2.22	1.12	0.26	0	0.53	0.26	0	0.53	0.19
			Max. Inv.	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80	10.80
			Avg. Inv.	7.74	7.74	7.38	7.02	7.74	7.38	7.02	6.66	6.30	7.02	6.66	6.30	6.48
			Cost	22.64	22.64	20.72	19.23	22.64	20.72	19.23	18.58	19.07	19.23	18.58	19.07	19.51
			Price	15.00	19.50	21.00	22.50	19.50	21.00	22.50	24.00	25.50	22.50	24.00	25.50	30.00
			Profit	—7.64	—3.14	0.28	3.27	—3.14	0.28	3.27	5.42	6.43	3.27	5.42	6.43	10.49

TABLE III: Problem Matrix for Heating Fu

										Refinery Strategies									
										Inventory I					24	24	27	27	30
										Production A					10.5	13.5	10.5	13.5	10.5
Vectors	P_0	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	P_{10}	P_{11}	P_{12}	P_{13}	P_{14}	P_{15}	P_{16}	P_{17}	P_{18}
P_1	0.25	1	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....
P_2	0.25	....	1	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....
P_3	0.05	....	....	1	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....
P_4	0.05	....	....	....	1	....	....	....	....	....	....	....	....	....	....	....	....	....	....
P_5	0.05	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	....	....	....
P_6	0.05	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	....	....
P_7	0.05	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	....
P_8	0.05	....	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....
P_9	0.125	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....
P_{10}	0.125	....	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....
P_{11}	0.025	....	....	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....	....
P_{12}	0.025	....	....	....	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....
P_{13}	1.0	....	....	....	....	....	....	....	....	....	....	....	....	1	....	....	....	....	....
P_{14}	0	....	....	....	....	....	....	....	....	....	....	....	....	....	1	....	....	....	....
P_{15}	0	....	....	....	....	....	....	....	....	....	....	....	....	....	....	1	....	....	....
P_{16}	0	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	1	....	....
P_{17}	0	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	1	....
P_{18}	0	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	1
P_{19}	0	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....
P_{-z}	—M	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....

TABLE IV: Optimal Solution of the Heating Fu

										Refinery Strategies									
										Inventory I					24	24	27	27	30
										Production A					10.5	13.5	10.5	13.5	10.5
Vectors	P_0	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	P_{10}	P_{11}	P_{12}	P_{13}	P_{14}	P_{15}	P_{16}	P_{17}	P_{18}
P_{30}	0.050	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	....
P_2	0.100	....	1	....	....	....	—1	—1	—1	....	....	....	....	....	....	....	....	....	....
P_{24}	0.025	....	....	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....	....
P_{29}	0.050	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	....	....
P_5	0.050	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....	....	....	....
P_{22}	0.150	1	....	—1	—1	....	....	....	....	....	....	....	....	....	....	....	....	....	....
P_{12}	0.025	....	....	....	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....
P_{32}	0.050	....	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....
P_{33}	3.786	—0.61	....	—7.49	—3.45	....	—5.23	—1.78	—0.92	—3.45	....	—4.66	....	5.43	....	....	—1	....	....
P_{10}	0.125	....	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....	....	....
P_{21}	0.050	....	....	....	1	....	....	....	....	....	....	....	....	....	....	....	....	....	....
P_{17}	0.1635	2.38	....	0.43	—0.03	....	—2.39	—2.39	—4.25	....	....	....	....	....	....	....	—1	1	....
P_{18}	0.2517	1.54	....	....	—0.46	....	—3.69	—2.35	—4.21	—1.30	....	—2.21	....	0.62	....	....	—1	....	1
P_{14}	2.919	—2.39	....	—2.38	1.21	....	3.07	1.86	1.86	1.21	....	0.57	....	3.07	1	....	—1	....	....
P_{15}	0.965	....	....	—0.03	1.21	....	1.21	....	....	1.21	....	0.57	....	0.68	....	1	—1	....	....
P_{19}	1.8215	2.38	....	0.43	—0.03	....	—5.23	—3.93	—8.14	—0.46	....	—1.24	....	2.16	....	....	—1	....	....
P_{26}	0.475	—1	....	....	....	....	—1	—1	—1	—1	....	....	....	1	....	....	....	....	....
P_{20}	0.050	....	....	1	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....
P_{25}	0.100	....	....	....	....	....	....	....	....	1	....	—1	....	....	....	....	....	....	....
P_{-z}	—3.786	0.61	....	7.49	3.45	....	5.23	1.78	0.92	3.45	....	4.66	....	M—5.43	....	....	1	....	....

ventory and Production Rate Problem

	Nature's Strategies														
	16.5 22.5	19.5 19.5	19.5 22.5	19.5 25.5	22.5 16.5	22.5 19.5	22.5 22.5	22.5 25.5	22.5 28.5	25.5 19.5	25.5 22.5	25.5 25.5	28.5 22.5	Requirement A Requirement B	
ectors	P_{20}	P_{21}	P_{22}	P_{23}	P_{24}	P_{25}	P_{26}	P_{27}	P_{28}	P_{29}	P_{30}	P_{31}	P_{32}	P_{33}	P_{34}
P_1	1	1	1	1	....	....	....	....	....	....	....	....	....	....	....
P_2	....	....	....	....	....	....	....	....	....	1	1	1	1	....	....
P_3	1	....	....	....	....	....	....	....	....	....	....	....	....	....	....
P_4	....	1	....	....	....	....	....	....	....	....	....	....	....	....	....
P_5	....	....	....	1	....	....	....	....	....	....	....	....	....	....	....
P_6	....	....	....	....	....	....	....	....	....	1	....	....	....	....	....
P_7	....	....	....	....	....	....	....	....	....	....	1	....	....	....	....
P_8	....	....	....	....	....	....	....	....	....	....	....	....	1	....	....
P_9	....	....	....	....	1	1	....	....	....	....	....	....	....	....	....
P_{10}	....	....	....	....	....	....	....	1	1	....	....	....	....	....	....
P_{11}	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....
P_{12}	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....
P_{13}	1	1	1	1	1	1	1	1	1	1	1	1	1	....	....
P_{14}	—0.97	—0.52	4.14	7.59	—7.53	—2.30	2.36	5.81	7.92	—5.94	—1.28	2.17	—0.42	—1	1
P_{15}	—3.32	—0.52	4.14	7.59	—5.14	0.09	4.75	8.20	10.31	—1.69	2.97	6.42	3.83	—1	1
P_{16}	—2.67	1.37	4.82	6.97	—2.68	1.98	5.43	7.54	8.18	0.20	3.65	5.76	4.51	—1	1
P_{17}	—5.48	—0.98	2.44	5.43	—2.68	1.98	5.43	7.58	8.59	2.59	6.04	8.15	8.76	—1	1
P_{18}	—4.83	—0.33	2.66	4.81	0.21	2.66	4.81	5.82	6.02	3.27	5.38	6.02	8.10	—1	1
P_{19}	—7.64	—3.14	0.28	3.27	—3.14	0.28	3.27	5.42	6.43	3.27	5.42	6.43	10.49	—1	1
P_{-z}	—M	—M	—M	—M	—M	—M	—M	—M	—M	—M	—M	—M	—M	1	—1

ventory and Production Rate Problem

	Nature's Strategies														
	16.5 22.5	19.5 19.5	19.5 22.5	19.5 25.5	22.5 16.5	22.5 19.5	22.5 22.5	22.5 25.5	22.5 28.5	25.5 19.5	25.5 22.5	25.5 25.5	28.5 22.5	Requirement A Requirement B	
ectors	P_{20}	P_{21}	P_{22}	P_{23}	P_{24}	P_{25}	P_{26}	P_{27}	P_{28}	P_{29}	P_{30}	P_{31}	P_{32}	P_{33}	P_{34}
P_{30}	....	....	....	....	....	....	....	....	....	....	1	....	....	....	....
P_2	....	....	....	....	....	....	....	....	....	....	....	1	....	....	....
P_{24}	....	....	....	....	1	....	....	....	....	....	....	....	....	....	....
P_{29}	....	....	....	....	....	....	....	....	....	1	....	....	....	....	....
P_5	....	....	....	1	....	....	....	....	....	....	....	....	....	....	....
P_{22}	....	....	1	1	....	....	....	....	....	....	....	....	....	....	....
P_{12}	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....
P_{32}	....	....	....	....	....	....	....	....	....	....	....	....	1	....	....
P_{33}	....	....	....	—2.15	....	....	....	—2.11	—2.75	....	....	—0.33	....	1	—1
P_{10}	....	....	....	....	....	....	....	....	1	....	....	....	....	....	....
P_{21}	....	1	....	....	....	....	....	....	....	....	....	....	....	....	....
P_{17}	....	....	....	0.84	....	....	....	0.04	—2.75	....	....	2.39	....	....	....
P_{18}	....	....	....	....	....	....	....	—1.10	0.41	....	....	0.88	....	....	....
P_{14}	....	....	....	1.30	....	....	....	1.34	—1.54	....	....	—0.52	....	....	....
P_{15}	....	....	....	1.30	....	....	....	1.34	2.81	....	....	1.34	....	....	....
P_{19}	....	....	....	0.84	....	....	....	0.04	2.81	....	....	2.83	....	....	....
P_{26}	....	....	....	....	....	....	1	1	0.41	....	....	1	....	....	....
P_{20}	1	....	....	....	....	....	....	....	....	....	....	....	....	....	....
P_{25}	....	....	....	....	....	1	....	....	....	....	....	....	....	....	....
P_{-z}	....	....	....	2.15	....	....	....	2.11	2.75	....	....	0.33	....	....	1

able profit for decreasing inventory 1 from 27 to 24 would be 0.965. On the other hand, the decrease in probable profit for increasing inventory 1 from 27 to 30 would be 0.252. It is necessary, of course, to study other cases and other factors in order to determine the optimum operation.

The selection of nature's strategies is made by the simplex method to inflict the greatest loss on the refinery operations within the limits of the constraints. This solution selected the following distribution of requirements:

Nature's Strategy	Req. A	Req. B	Total Req.	Probability
x_{20}	16.5	23.5	39	0.05
x_{21}	19.5	19.5	39	0.05
x_{22}	19.5	22.5	42	0.15
x_{23}	19.5	25.5	45	0
x_{24}	22.5	16.5	39	0.025
x_{25}	22.5	19.5	42	0.10
x_{26}	22.5	22.5	45	0.475
x_{27}	22.5	25.5	48	0
x_{28}	22.5	28.5	51	0
x_{29}	25.5	19.5	45	0.05
x_{30}	25.5	22.5	48	0.05
x_{31}	25.5	25.5	51	0
x_{32}	28.5	22.5	51	0.05
				1.00

It is concluded that this distribution is a satisfactory representation of past occurrences. However, since the strategies of nature which were omitted in this solution would tend to increase refiner's profit, the profit of 3.786 calculated by the linear programming solution of the constrained game must be considered somewhat pessimistic. This is, of course, a conservative method of planning operations.

CHAPTER 7

Planning an Experimental Program*

Introduction

In order to show the use of the constrained game theory, a simplified example has been chosen which represents alternate methods of selecting an experimental program to solve a refinery problem. This example is hypothetical, and fictional costs and probabilities have been arbitrarily chosen. The study was suggested by the work of Thomas A. Elkins on geophysical exploration.[4]

Statement of Problem

A refinery has run into difficulty with the color of the lube stock. It has been necessary to clay filter this stock at an additional cost to the refining operation of $1,000,000. A suggestion has been made for a modification in the extraction column which may eliminate the clay filtering. The modification will cost $100,000. There are several alternate schemes that the refiner can follow in deciding to make this plant modification. These are:

I. He can make the modification without any preliminary experimental testing.

II. He can perform a bench scale test and install the modification if the bench scale test is successful.

III. He can perform a pilot plant test and install the modification if the pilot scale test is successful.

IV. He can follow scheme III only if satisfactory results have been obtained on the bench scale test.

The possible results that can be obtained (the strategies of nature) are:

1. The modification will not clean up the lube stock and is indicated to be unsatisfactory by both the bench and pilot tests.

2. The modification will not clean up the lube stock and is indicated to be unsatisfactory by the pilot test, although satisfactory results are indicated by the bench test.

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*This chapter was prepared by Mr. M. A. Efroymson, Process Research Division, Esso Research and Engineering Company.

3. The modification will not clean up the lube stock, although satisfactory results are indicated by both the bench and pilot tests.
4. The modification is successful in the plant and is indicated to be satisfactory by both bench and pilot tests.
5. The modification is successful in the plant and is indicated to be satisfactory by the pilot plant test, although the bench test indicates it is unsatisfactory.
6. The modification is successful in the plant, although both the bench and pilot plants indicate it is unsatisfactory.

The payoff is then as below:

Nature	1	2	3	4	5	6
Refinery I	$-B$	$-B$	$-B$	$A-B$	$A-B$	$A-B$
Refinery II	$-C$	$-B-C$	$-B-C$	$A-B-C$	$-C$	$-C$
Refinery III	$-D$	$-D$	$-B-D$	$A-B-D$	$A-B-D$	$-D$
Refinery IV	$-C$	$-C-D$	$-B-C-D$	$A-B-C-D$	$-C$	$-C$

where
A = value of oil clean-up
B = cost of plant modification
C = cost of bench test
D = cost of pilot test.

We can assign values to these as follows:

(a) A successful lube oil clean-up is worth \$1,000,000 (A).
(b) The plant modification will cost \$100,000 (B).
(c) The bench test will cost \$5,000 (C).
(d) The pilot test will cost \$35,000 (D).

The payoff is then as below (in hundreds of thousands of dollars).

Nature	1	2	3	4	5	6
Refinery I	−0.1	−0.1	−0.1	0.9	0.9	0.9
Refinery II	−0.005	−0.105	−0.105	0.895	−0.005	−0.005
Refinery III	−0.035	−0.035	−0.135	0.865	0.865	−0.035
Refinery IV	−0.005	−0.040	−0.140	0.860	−0.005	−0.005

Discussion

If we assign x_1 to x_6 as the percentage of times nature employs strategies 1 through 6, then the payoff for each of the refiner's strategies can be written.

Profit for refiner's strategy

$$\begin{aligned}
\text{I} = Q_1 &= -0.1x_1 - 0.1x_2 - 0.1x_3 + 0.9x_4 + 0.9x_5 + 0.9x_6 \\
\text{II} = Q_2 &= -0.005x_1 - 0.105x_2 - 0.105x_3 + 0.895x_4 - 0.005x_5 - 0.005x_6 \\
\text{III} = Q_3 &= -0.035x_1 - 0.035x_2 - 0.135x_3 + 0.865x_4 + 0.865x_5 - 0.035x_6 \\
\text{IV} = Q_4 &= -0.005x_1 - 0.040x_2 - 0.140x_3 + 0.860x_4 - 0.005x_5 - 0.005x_6.
\end{aligned}$$

The refiner will choose the maximum Q obtained in these equations above.

If Q_0 = maximum Q, then $Q_1 - Q_0 \leq 0$; $Q_2 - Q_0 \leq 0$; $Q_3 - Q_0 \leq 0$; $Q_4 - Q_0 \leq 0$. Therefore the first equation above can be written

$$-0.1x_1 - 0.1x_2 - 0.1x_3 + 0.9x_4 + 0.9x_5 + 0.9x_6 - Q_0 \leq 0.$$

The other equations will have a similar form.

Since Q_0 may be either positive (a profit) or negative (a loss), it is desirable to define Q_0 as the difference of two non-negative numbers, i.e., $Q_0 = x_7 - x_8$, where x_8 is zero when Q_0 is positive and where x_7 is zero when Q_0 is negative.

The above equation can then be written

$$-0.1x_1 - 0.1x_2 - 0.1x_3 + 0.9x_4 + 0.9x_5 + 0.9x_6 - x_7 + x_8 \leq 0$$

or

$$-0.1x_1 - 0.1x_2 - 0.1x_3 + 0.9x_4 + 0.9x_5 + 0.9x_6 - x_7 + x_8 + x_9 = 0$$

where x_9 is zero or a positive value which changes the inequality to an equality; x_9 is called a slack variable, which is defined as a non-negative value that can be inserted into an inequation, thus making an equation.

The equation now states that the profit for scheme I

$$(-0.1x_1 - 0.1x_2 - 0.1x_3 + 0.9x_4 + 0.9x_5 + 0.9x_6)$$

minus the profit for the most profitable scheme $(x_7 - x_8)$ plus a slack value (x_9) which is zero or a positive value equals zero.

The other strategies for the refiner can be written as follows:

$$\begin{aligned}
&-0.005x_1 - 0.105x_2 - 0.105x_3 + 0.895x_4 - 0.005x_5 - 0.005x_6 - x_7 + x_8 + x_{10} = 0 \\
&-0.035x_1 - 0.035x_2 - 0.135x_3 + 0.865x_4 + 0.865x_5 - 0.035x_6 - x_7 + x_8 + x_{11} = 0 \\
&-0.005x_1 - 0.040x_2 - 0.140x_4 + 0.860x_4 - 0.005x_5 - 0.005x_6 - x_7 + x_8 + x_{12} = 0
\end{aligned}$$

where x_{10}, x_{11}, and x_{12} are other slack variables.

The limiting probabilities on the strategies of nature, x_1 to x_6, are obtained from our past experience with problems of this type. If they are stated as inequalities,

they should be stated, if possible, so that they establish a minimum value for favorable results or a maximum value for unfavorable results. In this particular problem, the most favorable results are those which are actually tested in the plant and are successful, and the most unfavorable are those which are actually tested in the plant and are unsuccessful. This can be stated in another way, namely, that we want the number of successful plant modifications to be as great as possible, while keeping the total number of plant modifications as small as possible.

These limiting probabilities on the successful plant modifications can be stated as follows (based on past experience):

(a) If we used no preliminary testing, there is a 10 per cent *or more* chance that the plant modification will clean up the oil.

(b) If the plant modification would clean up the oil, 70 per cent* *or more* of the time the bench test will be indicative, and we will install it in the plant.

(c) If the plant modification would clean up the oil, 80 per cent* *or more* of the time the pilot test will be indicative, and we will install it in the plant.

These restrictions can be stated algebraically as follows:

(a)	$x_4 + x_5 + x_6$	$\geqq 10\%$	or $x_1 + x_2 + x_3$	$\leqq 90\%$
(b)	$x_4/(x_4 + x_5 + x_6)$	$\geqq 70\%$	or $-0.3x_4 + 0.7x_5 + 0.7x_6$	$\leqq 0$
(c)	$(x_4 + x_5)/(x_4 + x_5 + x_6)$	$\geqq 80\%$	or $-0.2x_4 - 0.2x_5 + 0.8x_6$	$\leqq 0.$

The limiting probabilities on the total number of plant modifications (again, based on past experience) are:

(d) If we used no preliminary testing, we would install the plant modification 100 per cent of the time.

(e) If we use the bench test, the test will have sufficient screening power that we will install the plant modification only 50 per cent *or less* of the time.

(f) If we use the pilot test, the test will have sufficient screening power that we will install the plant modification only 25 per cent *or less* of the time.

These restrictions can be stated algebraically as follows:

(d)	$x_1 + x_2 + x_3 + x_4 + x_5 + x_6$	$= 1$
(e)	$x_2 + x_3 + x_4$	$\leqq 50\%$
(f)	$x_3 + x_4 + x_5$	$\leqq 25\%$

These six equations can now be written

*Note: These values are used only to illustrate the mathematical technique and are not necessarily indicative of the sensitivity of this kind of experimental work.

(a) $x_1 + x_2 + x_3 + x_{13} = 0.9$

(b) $-0.3x_4 + 0.7x_5 + 0.7x_6 + x_{16} = 0$

(c) $-0.2x_4 - 0.2x_5 + 0.8x_6 + x_{17} = 0$

(d) $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_{18} = 1$

(e) $x_2 + x_3 + x_4 + x_{14} = 0.5$

(f) $x_3 + x_4 + x_5 + x_{15} = 0.25$

where x_{13} to x_{17} are slack variables and x_{18} is an artificial variable. An artificial variable must have a value of zero in any optimal solution.

A solution is obtained by solving for the x values which satisfy the six equations above together with the four equations on page 49, which are written in matrix form as Tableau I in Table 1, page 52. We obtain an optimum solution by maximizing the function $-\lambda - M\,x_{18}$ where $\lambda = x_7 - x_8$.

The first, second, and last tableaus are shown on Table 1. The remaining steps in the calculation can be inferred from Table 2, page 52.

From the last tableau, the expected gain from this proposal is $23,500. The optimum strategy for the refiner is P_{12} (performing both bench and pilot tests) with the optimum strategy for nature being (0.47, 0.26, 0.17, 0.07, 0.01, 0.02).

These are average values based on the probability of each occurrence. As shown in the payoff matrix on page 48, any one particular effort may cost as much as $140,000 (nature strategy 3), or may return as much as $860,000 (nature strategy 4). However, if we made enough decisions of this kind so that the proportion of successful and unsuccessful ventures would average out according to the law of probability, we would realize an average profit of $23,500.

A more complete analysis of the economics of the alternate refinery schemes is shown in the table below:

ECONOMICS OF ALTERNATE REFINERY SCHEMES

Refinery Scheme	Average Cost			Average Income	Average Profit	Percent Return
	Bench and Pilot Tests	Plant Modification	Total			
No testing	—	100,000	100,000	100,000	—	—
Bench testing	5,000	50,000	55,000	70,000	15,000	27.2
Pilot testing	35,000	25,000	60,000	80,000	20,000	33.3
Bench and pilot testing	23,500	24,000	47,500	70,000	23,500	49.4

The use of bench and pilot tests shows the greatest profit ($23,500) and the

TABLE 1—SIMPLEX CALCULATION OF CONSTRAINED GAME THEORY PROBLEM

		P_0	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	P_{10}	P_{11}	P_{12}	P_{13}	P_{14}	P_{15}	P_{16}	P_{17}	P_{18}
									−1.0	1.0										−M
FIRST TABLEAU			↓																	
	P_9	....	−0.100	−0.100	−0.100	0.900	0.900	0.900	−1.0	1.0	1.0	....	....	....	....	....	....	....	....	....
	P_{10}	....	−0.005	−0.105	−0.105	0.895	−0.005	−0.005	−1.0	1.0	....	1.0	....	....	....	....	....	....	....	....
	P_{11}	....	−0.035	−0.035	−0.135	0.865	0.865	−0.035	−1.0	1.0	....	....	1.0	....	....	....	....	....	....	....
	P_{12}	....	−0.005	−0.040	−0.140	0.860	−0.005	−0.005	−1.0	1.0	....	....	....	1.0	....	....	....	....	....	....
	P_{13}	→ 0.9	1.0	1.0	1.0	....	....	....	....	....	....	....	....	....	1.0	....	....	....	....	....
	P_{14}	0.5	....	1.0	1.0	1.0	....	....	....	....	....	....	....	....	....	1.0	....	....	....	....
	P_{15}	0.25	....	....	1.0	1.0	1.0	....	....	....	....	....	....	....	....	....	1.0	....	....	....
	P_{16}	....	....	....	....	−0.3	0.7	0.7	....	....	....	....	....	....	....	....	....	1.0	....	....
	P_{17}	....	....	....	....	−0.2	−0.2	0.8	....	....	....	....	....	....	....	....	....	....	1.0	....
	−M P_{18}	1.0	1.0	1.0	1.0	1.0	1.0	1.0	....	....	....	....	....	....	....	....	....	....	....	1.0
	$z_j - c_j$	−M	−M	−M	−M	−M	−M	−M	1.0	−1.0	....	....	....	....	....	....	....	....	....	....
SECOND TABLEAU						↓														
	P_9	0.09	....	....	....	0.900	0.900	0.900	−1.0	1.0	1.0	....	....	....	0.1	....	....	....	....	....
	P_{10}	→ 0.0045	....	−0.100	−0.100	0.895	−0.005	−0.005	−1.0	1.0	....	1.0	....	....	0.005	....	....	....	....	....
	P_{11}	0.0315	....	....	−0.100	0.865	0.865	−0.035	−1.0	1.0	....	....	1.0	....	0.035	....	....	....	....	....
	P_{12}	0.0045	....	−0.035	−0.135	0.860	−0.005	−0.005	−1.0	1.0	....	....	....	1.0	0.005	....	....	....	....	....
	P_1	0.9	1.0	1.0	1.0	....	....	....	....	....	....	....	....	....	1.0	....	....	....	....	....
	P_{14}	0.5	....	1.0	1.0	1.0	....	....	....	....	....	....	....	....	....	1.0	....	....	....	....
	P_{15}	0.25	....	....	1.0	1.0	1.0	....	....	....	....	....	....	....	....	....	1.0	....	....	....
	P_{16}	....	....	....	....	−0.3	0.7	0.7	....	....	....	....	....	....	....	....	....	1.0	....	....
	P_{17}	....	....	....	....	−0.2	−0.2	0.8	....	....	....	....	....	....	....	....	....	....	1.0	....
	P_{18}	0.1	....	....	....	1.0	1.0	1.0	....	....	....	....	....	....	−1.0	....	....	....	....	1.0
	$z_j - c_j$	−0.1M	....	....	....	−M	−M	−M	1.0	−1.0	....	....	....	....	M	....	....	....	....	....
NINTH TABLEAU																				
	P_9	0.0235	....	....	....	....	....	....	....	....	1.0	....	....	−1.0	0.29	−0.35	−0.1	−0.9	−0.1	−0.195
	P_4	0.07	....	....	....	1.0	....	....	....	....	....	....	....	....	−0.7	....	....	1.0	....	0.7
	P_{11}	0.0035	....	....	....	....	....	....	....	....	....	....	1.0	−1.0	0.09	−0.35	....	−0.9	0.9	....
	P_2	0.26	....	1.0	....	....	....	....	....	....	....	....	....	....	−0.10	....	−1.0	1.0	−1.0	0.1
	P_1	0.47	1.0	....	....	....	....	....	....	....	....	....	....	....	0.30	1.0	....	−1.0	....	−0.7
	P_{10}	0.0085	....	....	....	....	....	....	....	....	....	1.0	....	−1.0	−0.01	0.065	−0.1	0.1	−0.1	0.01
	P_3	0.17	....	....	1.0	....	....	....	....	....	....	....	....	....	0.8	−1.0	1.0	....	1.0	−0.8
	P_5	0.01	....	....	....	....	1.0	....	....	....	....	....	....	....	−0.1	....	....	1.0	−1.0	0.1
	P_6	0.02	....	....	....	....	....	1.0	....	....	....	....	....	....	−0.2	....	....	....	−0.1	0.2
	−1−P_7	0.0235	....	....	....	....	....	....	1.0	−1.0	....	....	....	−1.0	−0.71	−0.065	−0.1	−0.9	−0.1	0.705
	$z_j - c_j$	−0.0235	....	....	....	....	....	....	....	....	....	....	....	1.0	0.71	0.065	0.1	0.9	0.1	−0.705+M

TABLE 2

Tableau	I	II	III	IV	V	VI	VII	VIII	IX
Vector Inserted		1	4	7	5	3	6	2	10
Vector Removed	13	10	18	16	15	17	12	14	

lowest total cost ($47,500). It can be noted in all the cases that, compared to no testing, the value of experimental work, either bench scale or pilot plant, is related to the lower manufacturing costs (plant modification costs) rather than to any increase in income.

Practical Application

This example is too simplified to have any practical value; it is illustrative of a method of linear programming and could probably have been solved by conventional mathematics. The technique, however, can be extended to include all the factors which are considered in a realistic plan for experimental research, such as availability of research funds, availability of technical manpower, effect on other research projects competing for funds, manpower, and equipment, effect of time of completion on the value of the project, personnel morale, etc. When these other factors are considered, the problem becomes so complex that it is difficult to obtain an optimum solution by conventional means and the methods of linear programming will have the most merit.

CHAPTER 8

A Transportation Problem

Linear programming can also be applied to a minimum cost transportation problem. The general transportation model is described by George B. Dantzig in his "Simplex Method Applied to a Transportation Problem."[9] Koopmans and Dantzig state that a transportation problem involving 25 origins and 60 destinations was solved in 9 man days by hand computation techniques. High-speed computers could undoubtedly solve this problem with very little effort. Henderson and Schlaifer[14] also describe an interesting transportation problem solved by linear programming. Although the general simplex method can be used for the solution of the transportation problem, it is more convenient to use a special solution technique applicable only to transportation and allocation problems.

Herewith is a typical minimum-cost-transportation problem solved by a special method of linear programming, a study of which has been presented by Charnes and Cooper in *Management Science*[13]. In this example given supplies of No. 2 Heating Oil are available at five origin points to meet stated requirements at three destination terminals. The transportation costs for all 15 possible transfers are known. There are many possible solutions to this problem, and only one of them will be at the minimum cost.

Development of the Mathematical Model

Let X_j = quantity available at the jth supply point,
Y_i = quantity required at the ith destination,
A_{ij} = quantity transferred from j to i,
and C_{ij} = unit cost of this transfer.

The constraints are

$$\Sigma_j A_{ij} = Y_i \qquad \begin{cases} i = 1 \cdots m \\ j = 1 \cdots n \\ A_{ij} \geqq 0. \end{cases}$$

and

$$\Sigma_i A_{ij} = X_j$$

The objective is

$$\Sigma_i \Sigma_j C_{ij} A_{ij} = f(C) = \text{minimum}.$$

The general simplex form for this model is illustrated for a 3 by 3 problem as follows:

A_{11}	A_{12}	A_{13}	A_{21}	A_{22}	A_{23}	A_{31}	A_{32}	A_{33}	f(C)	Q
1	1	1	..	..	..	..	..	..	..	$= Y_1$
..	..	..	1	1	1	..	..	..	..	$= Y_2$
..	..	..	..	..	..	1	1	1	..	$= Y_3$
1	..	..	1	..	..	1	..	..	..	$= X_1$
..	1	..	..	1	..	..	1	..	..	$= X_2$
..	..	1	..	..	1	..	..	1	..	$= X_3$
$-C_{11}$	$-C_{12}$	$-C_{13}$	$-C_{21}$	$-C_{22}$	$-C_{23}$	$-C_{31}$	$-C_{32}$	$-C_{33}$	1	$= 0$
									f(C)	= minimum

The six equations for Y_i and X_j contain one redundancy, since $\Sigma_i Y_i = \Sigma_j X_j$. Thus any Y_i or X_j value can be derived from known values of the other five variables. Therefore one of the equations can be eliminated from the set. Generally $m + n - 1$ variables will appear in the basis. By eliminating equation X_3 and rearranging, a feasible solution may be obtained. If variables A_{11}, A_{12}, A_{22}, A_{23}, and A_{33} form a non-negative basis, the following solution results:

A_{11}	A_{12}	A_{13}	A_{21}	A_{22}	A_{23}	A_{31}	A_{32}	A_{33}	f(C)	Q
....	1	1	-1	....	....	-1	....	....	....	$Y_1 - X_1$
....	....	1	....	....	1	-1	-1	....	....	$Y_2 + Y_1 - X_2 - X_1$
....	....	....	....	....	....	1	1	1	....	Y_3
1	....	....	1	....	....	1	....	....	....	X_1
....	....	-1	....	1	....	1	1	....	....	$X_2 + X_1 - Y_1$
....	....	$-C'_{13}$	$-C'_{21}$	....	....	$-C'_{31}$	$-C'_{32}$	....	1	$f(C)$
		$= -C_{13}$	$= -C_{21}$	....	....	$= -C_{31}$	$= -C_{32}$	....	....	
....	....	$+C_{23} - C_{22}$	$+C_{22} - C_{12}$	....	....	$+C_{33} - C_{23}$	$+C_{33} - C_{23}$	....	....	
....	....	$+C_{12}$	$+C_{11}$	....	....	$+C_{22} - C_{12}$	$+C_{22}$	....	....	
....	....			....	....	$+C_{11}$		....	....	

where

$$f(C) = C_{11}X_1 + C_{12}(Y_1 - X_1) + C_{22}(X_2 + X_1 - Y_1) + C_{23}(Y_2 + Y_1 - X_2 - X_1) + C_{33}Y_3.$$

This solution may be shown as follows:

	j			
i	1	2	3	$\Sigma_i A_{ij}$
1	X_1	Y_1-X_1		Y_1
2		$X_2+X_1-Y_1$	$Y_2+Y_1-X_2-X_1$	Y_2
3			Y_3	Y_3
$\Sigma_i A_{ij}$	X_1	X^2	X^3	

Note $$X_3=Y_3+Y_2+Y_1-X_2-X_1.$$

The effect on $f(C)$ of substituting any A_{ij} not in the basis for an A_{ij} in the basis is shown by the new C'_{ij} values (negative coefficients in the $f(C)$ row). Thus, for substituting A_{13},

$$\Delta f(C)/\Delta A_{13} = -(-C'_{13}) = -(-C_{13}+C_{23}-C_{22}+C_{12}).$$

This means that if A_{13} is increased by one unit, A_{23} must be decreased by one unit, A_{22} increased by one unit, and A_{12} decreased by one unit in order to maintain the required balance as follows:

	j		
i	1	2	3
1		−1	+1
2		+1	−1
3			

These changes will change $f(C)$ accordingly by the original C_{ij} values. If any new C'_{ij} value is positive, a decrease in $f(C)$ can be obtained by substituting the corresponding. A_{ij}. As in the simplex method, the entering A_{ij} should be large enough so that one of the A_{ij} values in the basis will become zero.

A method has been developed then for setting up a feasible solution to the trans-

portation problem and reaching for an optimum by efficient substitution based upon the general simplex method of linear programming. It will be noted that only addition and subtraction are used in the transformations, since all coefficients of A_{ij} in the simplex matrix are $+1$ or -1.

The Heating Oil Transportation Problem

A homogeneous material, available at supply points *A, B, C, D,* etc. in given quantities, should be shipped to destinations *X, Y, Z,* etc. to meet given requirements at the smallest possible cost. All transportation costs between origins and destinations are given.

Supplies (No. 2 Heating Oil)

Point of Origin	*Amount Available* (Mbpd)	*Destination*	*Amount Required* (Mbpd)
A	10	*X*	40
B	20	*Y*	80
C	30	*Z*	120
D	80		
E	100		

Transportation Costs

Point to Point	*cpb*	*Point to Point*	*cpb*	*Point to Point*	*cpb*
A to *X*	57	*A* to *Y*	53	*A* to *Z*	58
B to *X*	60	*B* to *Y*	52	*B* to *Z*	63
C to *X*	55	*C* to *Y*	50	*C* to *Z*	61
D to *X*	54	*D* to *Y*	59	*D* to *Z*	56
E to *X*	62	*E* to *Y*	51	*E* to *Z*	64

STEP 1

Arrange table of costs above minimum cost.

	A	B	C	D	E
X	7	10	5	4	12
Y	3	2	0	9	1
Z	8	13	11	6	14

STEP 2

Arrange table of quantities to be transported.

	A	B	C	D	E	Total
X						40
Y						80
Z						120
Total	10	20	30	80	100	240

STEP 3

Select a basic feasible solution (No. 1), starting in box XA and working down to box ZE*

	A	B	C	D	E	Total
X	10	20	10			40
Y			20	60		80
Z				20	100	120
Total	10	20	30	80	100	240

The transportation cost for feasible solution No. 1 is \$143,800 per day. The cost calculations are given in Step 8.

It is now necessary to determine whether feasible solution No. 1 uses the minimum possible costs. This is done by determining the C'_{ij} for the vacant boxes (A_{ij} not in the basis).

STEP 4

Compute C'_{ij}

$$C'_{XD} = C_{XD} - C_{YD} + C_{YC} - C_{XC} = 4 - 9 + 0 - 5 = -10$$

$$C'_{XE} = C_{XE} - C_{ZE} + C_{ZD} - C_{YD} + C_{YC} - C_{XC} = 12 - 14 + 6 - 9 + 0 - 5 = -10$$

$$C'_{YA} = C_{YA} - C_{YC} + C_{XC} - C_{XA} = 3 - 0 + 5 - 7 = 1$$

$$C'_{YB} = C_{YB} - C_{YC} + C_{XC} - C_{XB} = 2 - 0 + 5 - 10 = -3$$

$$C'_{YE} = C_{YE} - C_{ZE} + C_{ZD} - C_{YD} = 1 - 14 + 6 - 9 = -16$$

*N.B. The number of elements filled in the boxes must equal $m + n - 1$ or the sum of X, Y, Z, etc. plus A, B, C, etc. minus 1; thus $3 + 5 - 1 = 7$. In some cases it might be necessary to add 0 in one or more boxes.

$$C'_{ZA} = C_{ZA} - C_{ZD} + C_{YD} - C_{YC} + C_{XC} - C_{XA} = 8 - 6 + 9 - 0 + 5 - 7 = 9$$
$$C'_{ZB} = C_{ZB} - C_{ZD} + C_{YD} - C_{YC} + C_{XC} - C_{XB} = 13 - 6 + 9 - 0 + 5 - 10 = 11$$
$$C'_{ZC} = C_{ZC} - C_{ZD} + C_{YD} - C_{YC} = 11 - 6 + 9 - 0 = 14$$

The cost factor C'_{YE} has the largest negative* value of -16, which is the change in $f(C)$ for each unit of YE which can be substituted in the basis. The maximum quantity of YE should be introduced into the basis, protecting non-negativity.

STEP 5

Determine feasible solution No. 2 by adding quantity θ in box YE and balancing case.

	A	B	C	D	E	Total
X	10	20	10			40
Y			20	$60-\theta$	θ	80
Z				$20+\theta$	$100-\theta$	120
Total	10	20	30	80	100	240

The quantity θ is the minimum value necessary to eliminate one of the elements† in basic solution No. 1, namely $\theta = 60$, and $YD = 60 - 60$ and is eliminated. The new feasible solution No. 2 becomes

	A	B	C	D	E	Total
X	10	20	10			40
Y			20		60	80
Z				80	40	120
Total	10	20	30	80	100	240

The transportation cost for feasible solution No. 2 is \$134,200 per day, as shown in Step 8.

Calculation of the cost factors shows that C'_{ZA} now has the largest negative value of -7, and ZA should next be introduced into the basis.

*N.B. In case of a tie for the largest negative value of C'_{ij}, the choice for substitution is arbitrary.

†N.B. If two of the elements should happen to be eliminated by this procedure, one of them must be restored as a 0. Special procedure is required as discussed in Koopman's *Activity Analysis of Production and Allocation.*(5)

STEP 6

Determine feasible solution No. 3 by adding quantity θ in box ZA and balancing case.

	A	B	C	D	E	Total
X	$10-\theta$	20	$10+\theta$			40
Y			$20-\theta$		$60+\theta$	80
Z	θ			80	$40-\theta$	120
Total	10	20	30	80	100	240

The minimum value of θ necessary to eliminate one of the elements in solution No. 2 is $\theta = 10$. The new feasible solution No. 3 becomes

	A	B	C	D	E	Total
X		20	20			40
Y			10		70	80
Z	10			80	30	120
Total	10	20	30	80	100	240

The transportation cost for feasible solution No. 3 is \$133,500 per day, as shown in Step 8.

Calculation of the cost factors shows that C'_{ZB} now has the largest negative value of -5, indicating that ZB should next be introduced into the basis.

STEP 7

Determine feasible solution No. 4 by adding quantity θ in box ZB and balancing case.

	A	B	C	D	E	Total
X		$20-\theta$	$20+\theta$			40
Y			$10-\theta$		$70+\theta$	80
Z	10	θ		80	$30-\theta$	120
Total	10	20	30	80	100	240

The minimum value of θ necessary to eliminate one of the elements in solution No. 3 is $\theta = 10$. The new feasible solution No. 4 becomes

	A	B	C	D	E	Total
X		10	30			40
Y					80	80
Z	10	10		80	20	120
Total	10	20	30	80	100	240

Calculation of the cost factors shows that none of the C'_{ij} values is negative, indicating that no further reduction in cost can be made by substitution of the A_{ij} values.

STEP 8

Determine the minimum cost. All cases are calculated to show the magnitude of the reduction.

CASE 1

	A	B	C	D	E			
X	10×7	$+ 20 \times 10$	$+ 10 \times 5$			$=$	320	$\text{Cost}_1 =$
Y			20×0	$+ 60 \times 9$		$=$	540	$(2380 + 50 \times 240)$
Z				20×6	$+ 100 \times 14$	$=$	1520	$1000/100 =$
							2380	\$143,800 pd

CASE 2

	A	B	C	D	E			
X	10×7	$+ 20 \times 10$	$+ 10 \times 5$			$=$	320	$\text{Cost}_2 =$
Y			20×0	$+$	60×1	$=$	60	$(1420 + 50 \times 240)$
Z				$80 \times 6 +$	40×14	$=$	1040	$1000/100 =$
							1420	\$134,200 pd

CASE 3

	A	B	C	D	E			
X		20×10	$+ 20 \times 5$			$=$	300	$\text{Cost}_3 =$
Y			10×0	$+$	70×1	$=$	70	$(1350 + 50 \times 240)$
Z	10×8			$+ 80 \times 6 +$	30×14	$=$	980	$1000/100 =$
							1350	\$133,500 pd

CASE 4

	A	B	C	D	E			
X		$10 \times 10 + 30 \times 5$				$=$	250	The minimum cost for
Y					80×1	$=$	80	this example becomes
Z	$10 \times 8 + 10 \times 13$			$+ 80 \times 6 + 20 \times 14$		$=$	970	$\text{Cost}_{\text{min}} =$
							1300	$(1300 + 50 \times 240)$
								$1000/100 =$
								\$133,000 pd

Note that the minimum-cost feasible solution Case 4 in this example uses the two highest transportation costs 64c for E to Z and 63c for B to Z and does not use the lowest transportation cost 50c for C to Y. No other minimum-cost feasible solution may be possible in this problem with other transportation arrangements since all C'_{ij} values are positive (negative coefficients in $f(C)$ row). The minimum cost for this example is \$133,000 per day.

CHAPTER 9

A Crude Allocation Problem

The transportation model of linear programming can be used to determine an arrangement for allocating available crudes to several refineries in order to obtain the maximum possible profit. An illustrative example is presented herewith. In this example, ten crudes are available in stated quantities from 10,000 to 30,000 barrels per day each, with an aggregate availability of 200,000 barrels per day. Three refineries X, Y, and Z have incremental operations with stated requirements totaling 180,000 barrels per day. Of the available crude, 20,000 barrels per day will not be used. One of the refineries has two incremental operations, X_1 and X_2, at different efficiency levels. The net profit or loss for each crude in each refinery operation is given. It is assumed that the crude evaluations will reflect the resulting product distribution from these incremental operations. However, if further debits are encountered in the solution because of lack of product quality or for transportation of surplus products, suitable corrections can be made in the crude evaluations and the problem reworked until a realistic solution is obtained. In large-scale problems, the calculations can be handled easily on high-speed computers.

The mathematical treatment used in this example is similar to the simplex method applied to a transportation problem described in Chapter 8. However, the crude allocation problem is worked to maximize the profit, whereas the transportation problem is worked to minimize the cost. A modified method of selecting the basic case is introduced. It it is believed that this method will save about one-half of the trials required for the solution. A new method is also suggested to maintain the required number of elements in the quantity matrix (necessary for the simplex method). It is believed that this method, also, will save calculations and will prevent degeneracy of the solution.

The following calculations are shown in considerable detail so that the method can be used for the solution of current problems.

Data Given

Crude evaluation, availability and requirement

Crude	a	b	c	d	e	f	g	h	i	j	Required M bpd
	(Profit or loss of each crude at each refinery cpb)										
Refinery											
X_1	−6	3	17	10	63	34	15	22	−2	15	30
X_2	−11	−7	−16	9	49	16	4	10	−8	8	40
Y	−7	3	16	13	60	25	12	19	4	13	50
Z	−1	0	13	3	48	15	7	17	9	3	60
N	0	0	0	0	0	0	0	0	0	0	20 Not Used
Available M bpd	30	30	20	20	10	20	20	10	30	10	200

Solution of Problem

Construct a basic feasible solution in the modified manner. Select the row and column containing the greatest profit from the table (63 cpb) and place this row X_1 and this column *e* as the first row and column of a new quantity table for the basic feasible solution.

	e										Total
X_1	10										30
Total	10										

Enter the totals of 30 for X_1 and 10 for *e*. Place the smaller of these two numbers in box X_1 *e*. Thus Crude *e* is all assigned to refinery X_1.

Since refinery X_1 requires additional crude, select the next highest profit operation. This is crude *f* with a profit of 34 cpb. The second column on the quantity table is *f*, and all of crude *f* is assigned to refinery X_1, satisfying its requirements.

	e	f								Total
X_1	10	20								30
Total	10	20								

A new refinery and a new crude must be selected, again on the basis of greatest profit. The profit table shows crude *h* at refinery *Y* to be the next best operation at 19 cpb. The desired information is placed in the quantity table as before.

	e	f	h							Total
X_1	10	20								30
Y			10							50
Total	10	20	10							

Additional crude supply is selected for refinery *Y* on the basis of greatest profit, namely, crudes *c* with 16 cpb and *d* with 13 cpb. This satisfies the crude requirement at refinery *Y*.

	e	f	h	c	d					Total
X_1	10	20								30
Y			10	20	20					50
Total	10	20	10	20	20					

A new refinery and a new crude must be selected. Crude *j* at refinery X_2 is the next best operation, with a profit of 8 cpb. Crudes *g* and *b* with profits of 4 cpb and —7 cpb are next in order of preference for refinery X_2.

	e	f	h	c	d	j	g	b			Total
X_1	10	20									30
Y			10	20	20						50
X_2						10	20	10			40
Total	10	20	10	20	20	10	20	30			

It will be noted that 20 M bpd of crude *b* is left unassigned until all the crudes available for the next refinery are considered. The remaining refinery is *Z*, and the remaining crudes in order of preference are *i, b,* and *a* with profits of 9, 0, and —1 cpb respectively. These assignments are made, and the leftover crude is placed in row *N*.

	e	f	h	c	d	j	g	b	i	a	Total
X_1	10	20									30
Y			10	20	20						50
X_2						10	20	10			40
Z								20	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

Since the total number of elements placed in the boxes is only 12 and the number of elements required to apply the simplex method is $n+m-1=10+5-1=14$, it is necessary to place 0 in two boxes. For best results, the selections should be made on the basis of highest profit* and they must "connect" the groups *e f* and *h c d* and *j g b i a*. An inspection of the profit table shows that *Ye* and X_2e satisfy these requirements. The modified basic feasible solution is now complete.

*Where the profit is equal for two or more competing elements, a definite order of preference must be established for all elements to be used in the calculations.

	e	f	h	c	d	j	g	b	i	a	Total
X_1	10	20									30
Y	0		10	20	20						50
X_2	0					10	20	10			40
Z								20	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

For convenience, the table of real profits is rearranged to conform with the basic feasible solution.

	e	f	h	c	d	j	g	b	i	a
X_1	63	34	22	17	10	15	15	3	−2	−6
Y	60	25	19	16	13	13	12	3	4	−7
X_2	49	16	10	−16	9	8	4	−7	−8	−11
Z	48	15	17	13	3	3	7	0	9	−1
N	0	0	0	0	0	0	0	0	0	0

The profit for the basic feasible solution is

	e	f	h	c	d	j	g	b	i	a	
X_1	630	+ 680									= 1310
Y			+ 190	+ 320	+ 260						= 770
X_2						+ 80	+ 80	−70			= 90
Z								0 + 270	− 10		= 260
											2430

$$\text{Profit}_1 = 2430(1000/100) = \$24{,}300 \text{ per day}$$

To determine whether this is a maximum profit feasible solution, it is necessary to calculate the C'_{ij} values (negative coefficients in the $f(C)$ row of the simplex matrix). Since it is desired to maximize the profit function $f(C)$, any A_{ij} with a positive C'_{ij} value should be introduced into the basis.

Since there are 36 vacant boxes in this problem [$(mn - m - n + 1)$ possible

A_{ij} values not in the basis], it is unnecessary to compute all of the C'_{ij}. Actually the C'_{ij} should be computed in turn until the first positive value appears, and then the corresponding A_{ij} should be substituted. New C'_{ij} can then be computed by continuing the same order of computation until no more substitution can be made with increased profit. This procedure will save materially on the amount of computing over the usual simplex procedure of choosing the largest C'_{ij} for each substitution. For this case

$$C'_{X2d} = C_{X2d} - C_{Yd} + C_{Ye} - C_{X2e} = 9 - 13 + 60 - 49 = 7,$$

indicating that X_2d should be introduced into the basis. Therefore, the quantity θ is inserted in the X_2d position and the new case is balanced.

	e	f	h	c	d	j	g	b	i	a	Total
X_1	10	20									30
Y	$0+\theta$		10	20	$20-\theta$						50
X_2	$0-\theta$				θ	10	20	10			40
Z								20	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

The value of θ is the smallest necessary to eliminate one of the existing elements. This is 0, and element $X_2e = 0—0$ is eliminated. The new feasible solution No. 2 becomes

	e	f	h	c	d	j	g	b	i	a	Total
X_1	10	20									30
Y	0		10	20	20						50
X_2					0	10	20	10			40
Z								20	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

The profit for feasible solution No. 2 is the same as for the basic feasible solution, namely,

$$\text{Profit}_2 = \$24{,}300 \text{ per day.}$$

Continuing the search for positive C'_{ij} reveals that

$$C'_{Yb} = C_{Yb} - C_{X2b} + C_{X2d} - C_{Yd} = 3 + 7 + 9 - 13 = 6.$$

Therefore, Yb should be introduced into the basis.

The required adjustment is made by inserting θ in box Yb and balancing the case.

	e	f	h	c	d	i	g	b	i	a	Total
X_1	10	20									30
Y	0		10	20	$20-\theta$			θ			50
X_2					$0+\theta$	10	20	$10-\theta$			40
Z								20	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

Feasible solution No. 3 becomes

	e	f	h	c	d	i	g	b	i	a	Total
X_1	10	20									30
Y	0		10	20	10			10			50
X_2					10	10	20				40
Z								20	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

The profit for feasible solution No. 3 is

	e	f	h	c	d	j	g	b	i	a	
X_1	630 + 680										= 1310
Y	0		+ 190	+ 320	+ 130		+ 30				= 670
X_2					+ 90	+ 80	+ 80				= 250
Z							+ 0	+ 270	− 10		= 260
											2490

$$\text{Profit}_3 = 2490(1000/100) = \$24{,}900 \text{ per day}$$

Continuing the search for positive C'_{ij} reveals that

$$C'_{X1g} = C_{X1g} - C_{X2g} + C_{X2d} - C_{Yd} + C_{Ye} - C_{X1e} = 15 - 4 + 9 - 13 + 60 - 63 = 4.$$

Therefore, X_1g should be introduced into the basis, and θ is inserted in box X_1g.

	e	f	h	c	d	j	g	b	i	a	Total
X_1	$10-\theta$	20					θ				30
Y	$0+\theta$		10	20	$10-\theta$			10			50
X_2					$10+\theta$	10	$20-\theta$				40
Z								20	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

Feasible solution No. 4 becomes

	e	f	h	c	d	j	g	b	i	a	Total
X_1	0	20					10				30
Y	10		10	20				10			50
X_2					20	10	10				40
Z								20	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

Note that 0 has been restored in box X_1e since this is the highest profit element which will connect the two groups of elements.

The profit for feasible solution No. 4 is

	e	f	h	c	d	j	g	b	i	a	
X_1	0 + 680						+ 150				= 830
Y	600		+ 190 + 320					+ 30			= 1140
X_2					+ 180 + 80 +		40				= 300
Z								+	0 + 270 − 10		= 260
											2530

$$\text{Profit}_4 = 2530(1000/100) = \$25{,}300 \text{ per day}$$

The next A_{ij} to be introduced in the basis is X_2h since

$$C'_{X2h} = C_{X2h} - C_{X2b} + C_{Yb} - C_{Yh} = 17 - 0 + 13 - 19 = 1.$$

Therefore, θ is inserted in box Zh.

	e	f	h	c	d	j	g	b	i	a	Total
X_1	0	20					10				30
Y	10		10−θ	20				10+θ			50
X_2					20	10	10				40
Z			θ					20−θ	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

Feasible solution No. 5 becomes

	e	f	h	c	d	j	g	b	i	a	Total
X_1	0	20					10				30
Y	10			20				20			50
X_2					20	10	10				40
Z			10					10	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

The profit for feasible solution No. 5 is

	e	f	h	c	d	j	g	b	i	a		
X_1	0	+ 680					+ 150				=	830
Y	600			+ 320				+ 60			=	980
X_2					+ 180	+ 80	+ 40				=	300
Z			+ 170					+	0 + 270	− 10	=	430
												2540

$$\text{Profit}_5 = 2540(1000/100) = \$25{,}400 \text{ per day}$$

The search for positive C'_{ij} at this time requires the calculation of all 36 values, since no positive values can be found. This is therefore an optimal solution. However, there are two C'_{ij} values which are zero, namely,

$$C'_{Yg} = C_{Yg} - C_{Xlg} + C_{Xle} - C_{Ye} = 12 - 15 + 63 - 60 = 0$$

and

$$C'_{Zc} = C_{Zc} - C_{Zb} + C_{Yb} - C_{Yc} = 13 - 0 + 3 - 16 = 0$$

The substitution of *Yg* or *Zc* will result in different but equally profitable solutions known as alternate optima. These alternate solutions are shown as follows:

alternate solution No. 5A

	e	f	h	c	d	j	g	b	i	a	Total
X_1	10	20									30
Y	0			20			10	20			50
X_2					20	10	10				40
Z			10					10	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

and alternate solution No. 5B.

	e	f	h	c	d	j	g	b	i	a	Total
X_1	0	20					10				30
Y	10			10				30			50
X_2					20	10	10				40
Z			10	10					30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

Any combination of these solutions would also be at maximal profit; for example, an equal mixture of all three solutions would be:

	e	f	h	c	d	j	g	b	i	a	Total
X_1	3	20					7				30
Y	7			17			3	23			50
X_2					20	10	10				40
Z			10	3				7	30	10	60
N										20	20
Total	10	20	10	20	20	10	20	30	30	30	200

The selection of the optimal crude allocation, in this case, can be made with some flexibility, possibly to improve some other criterion such as service or good will. The profit will remain at $25,400 per day with any optimal solution.

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